

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 10

Biology and Science & Tech.

Solution

English



Khan Global Studies

Most Trusted Learning Platform

BPSC TEST-10 (SOLUTION)

Q 1. Ans. C

Explanation:

- The adrenal medulla is the central anatomical portion of the adrenal gland. It secretes two primary hormones known as **catecholamines**. These are **adrenaline** and **noradrenaline**. Adrenaline is also scientifically referred to as **epinephrine**. These hormones are released rapidly during periods of acute stress. They are also secreted during emergency situations. They are commonly known as emergency hormones or the hormones of "Fight or Flight." These chemicals increase the rate and strength of heart contractions. They raise the rate of respiration in the body. They also stimulate the breakdown of glycogen into glucose to provide immediate energy.
- **Cortisol**: It is a steroid hormone produced by the adrenal cortex. It is the primary glucocorticoid in the human body. It regulates carbohydrate metabolism. It also produces anti-inflammatory effects. It does not initiate the rapid emergency response.
- **Aldosterone**: It is a mineralocorticoid secreted by the outer adrenal cortex. It acts on the renal tubules of the kidneys. It promotes the reabsorption of sodium and water. It maintains blood pressure and electrolyte balance. It is not involved in the immediate fight-or-flight reaction.
- **Glucagon**: It is a peptide hormone secreted by the alpha cells of the pancreas. It acts primarily on the liver to increase blood sugar levels. It does not originate from the adrenal medulla. It is not classified as an emergency catecholamine.

Q 2. Ans. C

Explanation:

- **Sweet Potato** is a modification of an adventitious root that has become swollen with stored food. It does not possess nodes, internodes, or buds.
- **Potato** is an underground stem modification called a tuber. The "eyes" of a potato are actually stem nodes containing axillary buds.
- **Onion** is a modified underground stem called a bulb. The stem itself is highly reduced and disc-like at the base, while the fleshy, edible layers are actually modified leaves storing food.
- **Ginger** is an underground stem called a rhizome. It has distinct nodes, internodes, and scaly leaves.

• Additional Facts:

- ✓ **Golden Rule of Stems**: The botanical feature that separates an underground stem from a root is the presence of nodes and internodes. Roots never possess these structures.
- ✓ **Taproot Modifications**: Other common examples of root modifications storing food include the Carrot, Turnip, and Radish (which are modified taproots, unlike the sweet potato, which is an adventitious root).
- ✓ **Vegetative Propagation**: Because potatoes, ginger, and onions are stems with buds, they can be directly used to grow a new plant asexually (vegetative propagation), which is a common agricultural practice.

Q 3. Ans. D

Explanation:

- **Statement 1 is correct**: Insulin is a peptide hormone. It is secreted by the beta-cells of the Islets of Langerhans in the pancreas. This hormone plays a central role in glucose homeostasis. It acts mainly on hepatocytes, adipocytes and muscle cells. It enhances cellular glucose uptake and utilisation. This process results in a rapid movement of glucose from the blood to the cells. Consequently, the blood glucose levels decrease. This physiological state of lowered blood sugar is known as hypoglycemia.
- **Statement 2 is correct**: Glucagon is a peptide hormone secreted by the alpha-cells of the pancreas. It acts as an antagonist to insulin. Glucagon is released into the bloodstream when blood glucose levels are low. It acts primarily on the liver cells. It stimulates the breakdown of stored glycogen into glucose. This biochemical process is called glycogenolysis. Glucagon also stimulates the process of gluconeogenesis. Both processes contribute to an increase in blood sugar levels. This state of elevated blood sugar is known as hyperglycemia.
- **Glucose Homeostasis**: The human body maintains a constant internal environment through the balanced action of these two hormones. High blood sugar triggers the secretion of insulin to promote storage and usage. Low blood sugar triggers the release of glucagon to mobilise stored energy. This feedback mechanism ensures that body cells receive a steady supply of fuel. A prolonged state of high blood sugar leads to a complex disorder called

diabetes mellitus. This condition is often associated with the loss of glucose through urine and the formation of harmful ketone bodies.

Q 4. Ans. B

Explanation:

- In a mango, the fruit wall (pericarp) is well-differentiated. The middle layer, known as the **mesocarp**, is the thick, fleshy, sweet, and juicy tissue that is commercially consumed.
- **Epicarp** is the thin, outermost skin or rind of the mango, which is generally not consumed.
- **Endocarp** in a drupe fruit is the hard, stony layer that immediately surrounds and protects the actual seed inside.
- **Thalamus** is the base of the flower. In some plants (like the apple or strawberry), the thalamus swells up to become the fleshy edible part, making them “false fruits.” A mango, however, is a “true fruit” developed directly from the ovary.
- Additional Facts:
 - ✓ While both mango and coconut are drupes, their edible parts are entirely different. In a coconut, the mesocarp is fibrous (yielding coir, not flesh), the endocarp is the hard brown shell, and the white edible part is the endosperm (the food reserve for the seed). In a coconut, the white edible flesh is the endosperm, and the coconut water/milk is the liquid endosperm.
 - ✓ Drupe fruit: Botanically, a drupe is a fleshy fruit that develops from a monocarpellary (single carpel), superior ovary, and typically contains only one seed. Examples: Mango, coconut, peach, plum, cherry, olive, almond, walnut, pistachio, coffee bean.
 - ✓ True vs. False Fruits: True fruits develop solely from the fertilised ovary (e.g., Mango, Grape). False fruits incorporate other floral parts, like the thalamus, into the fruit body (e.g., Apple, Pear, Cashew).

Q 5. Ans. C

Explanation:

- **Statement 1 is correct:** The ABO blood grouping system relies fundamentally on erythrocyte surface chemistry. It is specifically based on the physical presence or absence of two distinct surface antigens. These specific immune-triggering chemicals are designated as antigen A and antigen B.
- **Statement 2 is correct:** Individuals with the ‘O’ blood type completely lack both A and B antigens

on the surface of their red blood cells. Their systemic blood plasma continuously contains both natural anti-A and anti-B antibodies. This specific biochemical profile ensures their blood does not trigger an immune response when transfused into persons with other ABO blood types.

Q 6. Ans. C

Explanation:

- Web 3.0 is the next iteration of the internet, fundamentally built on decentralisation. It uses blockchain technology to remove central authorities (like big tech companies), giving users direct ownership and control over their own data and digital assets.
- Web 3.0 actively shifts away from centralised cloud servers (like AWS or Google Cloud) toward decentralised, peer-to-peer node networks.
- Web 1.0 (roughly 1990–2004), which consisted of basic, static pages where users could only consume information.
- Web 2.0 (the current web), defined by interactive, user-generated content (e.g., YouTube, Facebook), is controlled by centralised tech giants.
- **Additional facts:**
 - ✓ dApps (Decentralised Applications): Software applications that run on a blockchain or peer-to-peer network rather than a single computer or server.
 - ✓ Smart contracts: Self-executing contracts with the terms of the agreement directly written into lines of code, functioning autonomously on the blockchain.
 - ✓ Tokenomics: Web 3.0 heavily relies on cryptocurrencies and non-fungible tokens (NFTs) to facilitate transactions and prove digital ownership without a bank or intermediary.

Q 7. Ans. B

Explanation:

- **Erythroblastosis foetalis** is a severe haemolytic disease of the newborn. This condition occurs due to Rh antigen incompatibility between a pregnant mother and her developing foetus. It specifically develops when an Rh-negative mother carries an Rh-positive foetus. During the delivery of the first child, maternal blood may mix with foetal blood. The mother’s immune system subsequently produces specific anti-Rh antibodies within her blood plasma. In subsequent pregnancies, these

accumulated maternal antibodies can cross the placental barrier. These antibodies actively destroy the Rh-positive red blood cells of the new foetus. This rapid immunological destruction causes severe anaemia and jaundice in the newborn.

- **Haemophilia:** It is an inherited genetic bleeding disorder. It specifically impairs the body's physiological ability to form functional blood clots. This condition is a sex-linked recessive trait. It is entirely unrelated to immunological maternal-foetal blood group incompatibilities.
- **Sickle cell anaemia:** It is an inherited genetic condition affecting the physical shape of red blood cells. It is biologically classified as an autosomal recessive trait. It causes flexible red blood cells to become rigidly sickle-shaped under low oxygen tension. This physical distortion is caused by a specific point mutation in the beta-globin gene rather than maternal antibodies.
- **Thalassemia:** It is an inherited autosomal recessive blood disorder. It directly results in the abnormal or significantly reduced synthesis of vital haemoglobin chains. This genetic defect causes the formation of small, pale red blood cells and subsequent systemic anaemia. It is a quantitative genetic defect rather than an immunological reaction driven by transplacental antibodies.

Q 8. Ans. D

Explanation:

- **Reverse Transcription Polymerase Chain Reaction (RT-PCR)** is not a gene-editing technology. It is a highly sensitive laboratory technique used to detect and quantify specific RNA molecules (such as the RNA genome of the SARS-CoV-2 virus). It works by reverse-transcribing RNA into complementary DNA (cDNA) and then exponentially amplifying that DNA sequence. It is a diagnostic and amplification tool, not a genetic modification tool.
- **Additional facts:**
 - ✓ **CRISPR-Cas9** (Clustered Regularly Interspaced Short Palindromic Repeats)
 - **Mechanism:** The most revolutionary and widely adopted system. It utilises a customizable, synthesised guide RNA (gRNA) to direct the Cas9 endonuclease enzyme to a highly specific locus on the DNA strand, where it executes a precise cut.
 - **Significance:** Awarded the Nobel Prize in Chemistry in 2020, it is highly favoured

for its cost-effectiveness, speed, and ability to edit multiple genes simultaneously (multiplexing).

- ✓ **TALENs (Transcription Activator-Like Effector Nucleases)**
 - **Mechanism:** Utilises engineered restriction enzymes generated by fusing a specific DNA-binding domain (derived from *Xanthomonas* bacteria) to a DNA-cleaving domain.
 - **Significance:** While more difficult to engineer than CRISPR, TALENs offer an exceptionally high degree of precision and target binding specificity, making them highly valuable in clinical therapeutics and agricultural biotechnology.
- ✓ **ZFNs (Zinc Finger Nucleases)**
 - **Mechanism:** Represents the first generation of targeted genome editing tools. They are artificial restriction enzymes created by fusing a zinc finger DNA-binding domain (which recognises specific DNA triplets) to a DNA-cleavage domain (usually from the FokI restriction enzyme).
 - **Significance:** Historically significant as the pioneering programmable editing tool, though largely superseded by CRISPR and TALENs due to the complex and expensive protein-engineering required for each specific genomic target.
- ✓ **Homing Endonucleases (Meganucleases)**
 - **Mechanism:** Naturally occurring restriction enzymes are characterised by very large recognition sites (ranging from 12 to 40 base pairs). Because the sequence they recognise is so long, it typically occurs only once in any given genome.
 - ✓ **Significance:** They are the most precise of the naturally occurring nucleases and induce the lowest cellular toxicity. However, their strict reliance on fixed, naturally occurring recognition sequences makes them highly difficult to reprogram for novel, custom DNA targets.

Q 9. Ans. B

Explanation:

- **Vitamin K** acts as an essential cofactor for the enzyme gamma-glutamyl carboxylase, which is required for the synthesis of prothrombin (Factor II) and other clotting factors in the liver.

- The conversion of fibrinogen to fibrin is catalysed by the enzyme **thrombin**, not Vitamin K.
- Platelets are cellular fragments produced from megakaryocytes in the bone marrow; their production is regulated by **thrombopoietin**, not Vitamin K.
- **Heparin** is an anticoagulant that inhibits clotting; Vitamin K's role is the opposite, to facilitate the clotting process.
- **Additional Facts**
 - ✓ Specific clotting factors: Vitamin K is necessary for the synthesis of four major coagulation factors: Factor II (Prothrombin), VII, IX, and X.
 - ✓ Sources of Vitamin K:
 - Vitamin K1 (Phylloquinone): Primarily obtained from green leafy vegetables (e.g., spinach, kale).
 - Vitamin K2 (Menaquinone): Synthesised by symbiotic bacteria in the human colon (large intestine).
 - ✓ Significance: A deficiency in Vitamin K leads to an increased Prothrombin Time (PT), resulting in delayed clotting and excessive bleeding even from minor injuries.
 - ✓ Common blood-thinning medications like Warfarin work by interfering with the Vitamin K cycle, thereby reducing the liver's ability to produce clotting factors.
 - ✓ Beyond blood, Vitamin K is vital for bone health as it helps in the carboxylation of osteocalcin, a protein that binds calcium to the bone matrix.

Q 10. Ans. A

Explanation:

- The opening and closing of stomata are fundamentally driven by changes in the **turgor pressure** of guard cells. When water enters the guard cells, they become turgid and swell, causing the pore to open. When they lose water and become flaccid, the pore closes.
- **Additional Facts:**
 - ✓ **Potassium Ion Pump Theory:** The influx of water into guard cells is driven by an osmotic gradient. During the day, guard cells actively pump in potassium ions, which lowers their water potential and forces water to enter via endosmosis.
 - ✓ **Microfibril orientation:** The cellulose microfibrils in the cell walls of guard cells

are oriented radially (like a fan) rather than longitudinally. This specific structural adaptation makes it easier for the stoma to pull open when the cells swell.

- ✓ **Abscisic Acid (ABA):** Often referred to as the "stress hormone," ABA is synthesised by the plant during severe drought or water stress. It signals the guard cells to lose potassium and water, forcing the stomata to close and conserve moisture.
- ✓ **Shape of Guard Cells:** In dicotyledonous plants (dicots), guard cells are bean-shaped or kidney-shaped. However, in monocotyledonous plants (like grasses), the guard cells are distinctly dumbbell-shaped.

Q 11. Ans. A

Explanation:

- A chemical synapse transmits neural impulses through the release of specific neurotransmitters. This specific type of synapse is characterised by a distinct physical gap. This fluid-filled space separates the membrane of the presynaptic neuron from the membrane of the postsynaptic neuron. This specific anatomical space is called the **synaptic cleft**. Neurotransmitters must diffuse across this cleft to bind to specific receptors on the post-synaptic membrane.
- **Node of Ranvier:** It is a distinct microscopic gap. However, this gap is located between adjacent myelin sheaths along a single myelinated axon. It is not a junction between two separate neurons. It facilitates the rapid, jumping transmission of electrical impulses down the length of the axon.
- **Synaptic knob:** It is a swollen, bulb-like structure. It is located at the very terminal end of an axon branch. It is a solid physical cellular structure rather than a spatial gap. It houses the synaptic vesicles that physically store the chemical neurotransmitters prior to release.
- **Motor end plate:** It is a highly folded, specialised region of a muscle fibre membrane. It forms the receiving side of a neuromuscular junction. It is a physical membrane surface rather than the actual fluid-filled space between the communicating cells.

Q 12. Ans. C

Explanation:

- CMV is transmitted primarily by aphids (sap-sucking insects) in a "non-persistent" manner. This means the virus only stays on the aphid's

mouthparts (stylet) for a short time. When an infected aphid probes a healthy plant, the virus is immediately transmitted. Over 80 different species of aphids are capable of acting as vectors for CMV.

- While whiteflies are major agricultural pests, they are primarily vectors for a different class of major plant viruses, specifically Begomoviruses (such as the Cotton Leaf Curl Virus).
- Nematodes in the soil act as vectors for Nepoviruses and Tobraviruses, not CMV.
- Fungal zoospores transmit certain soil-borne viruses (like the Tobacco Necrosis Virus), but not CMV, which is transmitted above ground.
- **Additional facts:**
 - ✓ Unlike many viruses that target specific plant families, CMV infects over 1,200 plant species across 100 families, including cucumbers, melons, tomatoes, peppers, spinach, and many ornamentals.
 - ✓ CMV is an RNA virus (specifically, a positive-sense tripartite single-stranded RNA virus).
 - ✓ It causes characteristic “mosaic” or mottled patterns of light and dark green on leaves, stunted growth, and severely malformed fruits. In tomatoes, it famously causes a “shoestring” symptom where leaf blades fail to develop properly.

Q 13. Ans. D

Explanation:

- **Assertion (A) is false:** DNA profiling explicitly avoids coding regions (exons). Instead, it relies strictly on non-coding regions (introns or “junk DNA”) to identify individuals.
- **Reason (R) is true:** Roughly 99.9% of the human genome’s coding regions are identical in all humans because they contain the essential instructions for building basic human proteins. Because they do not vary, they are useless for distinguishing one suspect from another.
- **Additional facts:**
 - ✓ VNTRs and STRs: The non-coding regions contain repetitive DNA sequences. The number of times these sequences repeat varies drastically from person to person. These are called Variable Number Tandem Repeats (VNTRs) or Short Tandem Repeats (STRs).
 - ✓ High Polymorphism: It is the high degree of variation (polymorphism) in these VNTRs/STRs that creates a unique “fingerprint” for every individual (except identical twins).

- ✓ **Pioneer:** The technique of DNA fingerprinting was initially developed by British geneticist Alec Jeffreys in 1984.
- ✓ In India, Dr. Lalji Singh is considered the father of DNA fingerprinting.

Q 14. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The bone marrow is anatomically classified as a primary lymphoid organ. It is the main lymphoid tissue of the human body. All cellular components of the blood are produced within the bone marrow. This includes the initial biological formation of all lymphocytes.
- **Pair 2 is incorrectly matched:** The spleen is a secondary lymphoid organ. It is not a primary site for lymphocyte formation. The spleen contains mainly mature lymphocytes and phagocytes. It functions primarily as a systemic blood filter. It achieves this by trapping blood-borne pathogenic microorganisms.
- **Pair 3 is correctly matched:** The thymus is a primary lymphoid organ. It is a lobed anatomical structure located near the heart and beneath the breastbone. It provides the specific physiological micro-environment required for T-lymphocytes. Immature T-lymphocytes migrate here from the bone marrow to undergo their final maturation.
- **Pair 4 is incorrectly matched:** Lymph nodes do not trap microorganisms directly from the systemic bloodstream. They are small, solid structures situated at specific intervals along the lymphatic vessels. They specifically trap foreign antigens and microorganisms present in the interstitial tissue fluid and lymph. These trapped antigens subsequently activate the mature lymphocytes residing within the lymph node to trigger an immune response.

Q 15. Ans. B

Explanation:

- **Site-Directed Nucleases (SDNs)**, often colloquially referred to as “molecular scissors,” are engineered or naturally occurring enzymes that are programmed to recognise specific DNA sequences and induce targeted double-strand breaks (DSBs). Once the DNA is cut, the cell’s natural repair mechanisms kick in, allowing scientists to knock out, alter, or insert specific genes. All major gene-editing tools—including CRISPR-Cas9, TALENs, and ZFNs—are fundamentally types of SDNs.
- **Additional facts:**

- ✓ Genome editing via SDNs is classified into three categories based on the repair mechanism and the final genetic product:
 - SDN-1 (Gene Knockout / Silencing)
- Mechanism: The nuclease cuts the DNA, and the cell repairs the break naturally through Non-Homologous End Joining (NHEJ). This often results in small deletions or insertions (indels) that silence a specific target gene.
- Result: No foreign genetic material (from another species) is introduced.
- Regulatory Status: Recently, the Indian Government (MoEFCC) exempted SDN-1 from the stringent regulations of the Environment (Protection) Act, 1986, recognising that it mimics natural mutations.
 - SDN-2 (Targeted Nucleotide Editing)
- Mechanism: A targeted cut is made, but a small DNA template is provided to guide the repair process (Homology-Directed Repair - HDR), altering a few specific base pairs.
- Result: No foreign DNA is introduced; it merely edits the existing plant genome to express a desired trait.
- Regulatory Status: Exempted from strict GMO regulations in India alongside SDN-1.
 - SDN-3 (Transgenesis)
- Mechanism: The nuclease cuts the DNA, and a large template containing an entire foreign gene (often from a different species) is inserted into the break.
- Result: The creation of a classic Genetically Modified Organism (GMO) or transgenic plant (e.g., Bt Cotton, which contains a gene from the bacterium *Bacillus thuringiensis*).
- Regulatory Status: Strictly regulated under the EPA, 1989 rules, requiring clearance from the Genetic Engineering Appraisal Committee (GEAC).

Q 16. Ans. A

Explanation:

- The human stomach opens directly into the small intestine through the pyloric sphincter. The small intestine is anatomically divided into three distinct, sequential regions.
 - ✓ The first C-shaped region emerging from the stomach is the duodenum.
 - ✓ The duodenum is followed by a long, coiled middle portion called the jejunum.
 - ✓ The jejunum leads into a highly coiled terminal region called the ileum.
 - ✓ The ileum directly opens into the large intestine.

- ✓ The large intestine begins with a small blind sac called the caecum.
- ✓ The caecum continues upward into the colon.
- ✓ The colon is physically divided into ascending, transverse, descending, and sigmoid segments.
- ✓ The descending segment finally opens into the rectum.
- ✓ The rectum temporarily stores faecal matter prior to defaecation through the anus.
- The jejunum is strictly the middle portion of the small intestine. It does not anatomically connect to the stomach to receive acidic chyme.
- The ileum is strictly the terminal portion of the small intestine. It specifically serves as the physical bridge connecting the small intestine to the large intestine.
- The colon anatomically and functionally follows the caecum. The caecum is a distinct sac that hosts symbiotic micro-organisms. It receives undigested material directly from the ileum before passing it into the ascending colon.

Q 17. Ans. D

Explanation:

- Organ-on-a-chip (OoC) refers to a highly advanced, micro-scale, in vitro (outside the living body) system. It utilises microfluidic chips embedded with living human cells that are cultured in 3D structures using specialised polymers. These chips successfully mimic the complex physiological, mechanical, and biochemical microenvironments of actual human organs (like the lungs, heart, or liver), allowing researchers to observe how an organ reacts to certain drugs or diseases in real time.
- OoC technology is categorised under NAMs (New Approach Methods). These are innovative alternative testing strategies that aim to replace, reduce, or refine the use of laboratory animals in biological and medical research.

Q 18. Ans. C

Explanation:

- The Hypothalamus acts as the body's internal thermostat. It regulates body temperature, hunger, thirst, sleep cycles, and links the nervous system to the endocrine system via the pituitary gland to maintain overall homeostasis.
- Cerebellum is located at the back of the brain and is responsible for coordinating voluntary muscle movements, posture, and balance.
- Medulla oblongata is part of the brainstem and

controls autonomic, life-sustaining functions such as heart rate, breathing, and blood pressure.

- Thalamus acts primarily as a sensory relay station, transmitting sensory signals (except smell) to the cerebral cortex for processing.

Q 19. Ans. B

Explanation:

- **Quantum Superposition** is the ability of a quantum system to exist in multiple states at the exact same time until it is measured. This is what allows quantum computers to process massive numbers of possibilities simultaneously, unlike classical bits which are strictly binary (either 0 or 1).
- **Quantum Entanglement** refers to two or more qubits becoming intrinsically linked, so that the state of one instantly dictates the state of the other, regardless of the physical distance between them. It facilitates rapid data transfer within the system.
- **Quantum Tunnelling** is the phenomenon where a subatomic particle passes through a physical barrier it classically shouldn't be able to cross.
- **Quantum Decoherence** is the primary obstacle in quantum computing. It is the degradation of a qubit's quantum state back to a classical state due to minor environmental interference (like heat or electromagnetic waves).
- Additional facts:
 - ✓ Quantum Supremacy: The milestone where a quantum computer solves a problem that no classical supercomputer can solve in any feasible amount of time (e.g., Google's Sycamore processor in 2019).
 - ✓ Cryptography Threat: Quantum computers running Shor's Algorithm theoretically have the power to break modern RSA encryption, triggering a global push for "Post-Quantum Cryptography."
 - ✓ National Mission on Quantum Technologies & Applications (NMQTA): India launched this mission (approved in 2023 with a ₹6000+ crore budget) to develop indigenous quantum computers, secure communications, and quantum sensors.

Q 20. Ans. A

Explanation:

- **Statement 1 is correct:** Skeletal muscles are capable of generating force only through contraction. This contraction results in a pulling force on the attached bones. Muscles do not possess a

biological mechanism to actively push a bone away. Consequently, they must work in pairs to move a limb in opposite directions.

- **Statement 2 is incorrect:** During the flexion (bending) of the arm, the biceps muscle acts as the agonist or "prime mover." The biceps undergo active contraction to pull the forearm toward the shoulder. Simultaneously, the triceps muscle acts as the antagonist. The triceps must relax to allow the biceps to shorten and move the joint.

Additional facts:

- ✓ Antagonistic Pairs: This term refers to two muscles that exert opposing forces on the same joint. When one muscle contracts, the other must relax to facilitate movement.
- Agonist (prime mover): This is the muscle responsible for the primary movement of a joint. In arm extension (straightening), the triceps are the agonist.
- Antagonist: This muscle opposes the movement of the agonist. It must stay relaxed while the agonist contracts. In arm extension, the biceps act as the antagonist and relax.

Q 21. Ans. B

Explanation:

- **Assertion (A) is true:** The continuous water column in a tree, even one 100 meters tall, does not collapse or break under the force of gravity.
- **Reason (R) is true:** Transpiration genuinely creates a negative hydrostatic pressure (suction pull) in the xylem, which is the primary driving force for the ascent of sap.
- **Why (R) does not explain (A):** While (R) explains why the water moves upward, it does not explain why the column remains intact without breaking. In fact, the suction pull mentioned in (R) places the water column under severe physical stress. The actual reason the column does not break (explaining A) is due to the high tensile strength of water, which is derived from cohesion (mutual attraction between water molecules) and adhesion (attraction of water molecules to the lignocellulosic walls of the xylem).
- **Additional Facts:**
 - ✓ **Dixon and Joly (1894):** They proposed the "Cohesion-Tension Theory," which remains the most widely accepted model for water transport in tall plants.
 - ✓ **Capillarity:** Besides cohesion and adhesion, water rises in the narrow xylem vessels

(tracheids and vessels) due to surface tension, a phenomenon known as capillarity.

- ✓ **Transpiration Magnitude:** The transpiration-driven ascent of sap can pull water upwards at a speed of up to 15 meters per hour.
- ✓ **Cavitation/Embolism:** If the tension becomes too extreme (e.g., during severe drought or freezing), an air bubble can form in the xylem, breaking the water column. This is called cavitation, and plants have physiological mechanisms to bypass or repair these embolisms.

Q 22. Ans. B

Explanation:

- **Statement 1 is correct:** The intercellular material of cartilage is structurally solid. This material is also pliable. This specific biochemical composition allows the tissue to effectively resist mechanical compression.
- **Statement 2 is incorrect:** Chondrocytes are the specialised primary cells of cartilage. These cells are not freely suspended in a liquid medium. They are strictly enclosed within small microscopic cavities called lacunae. These lacunae are located deep within the solid matrix secreted by the chondrocytes themselves.
- **Statement 3 is incorrect:** Cartilage tissue is fundamentally avascular. It completely lacks a direct internal supply of blood vessels. Nutrients and oxygen must slowly diffuse through the solid ground substance to reach the chondrocytes. This absolute lack of direct blood flow results in extremely slow tissue repair and regeneration following a mechanical injury.
- **Statement 4 is correct:** Cartilage is anatomically present in the tip of the human nose. It forms the flexible structural framework of the outer ear joints. It is also strategically located between adjacent bones of the adult vertebral column to function as a vital shock absorber.

Q 23. Ans. B

Explanation:

- **Cashew** is a classic example of a **false fruit**. In the cashew plant, the fleshy, sweet, apple-like part that is often consumed locally (the cashew apple) actually develops from the pedicel (flower stalk) and the thalamus (receptacle), not the ovary. The actual “true fruit” is the kidney-shaped nut attached to the bottom of this fleshy structure.

- Banana is a **parthenocarpic fruit**, meaning it develops without fertilisation (hence it is seedless), but it is still a “true fruit” because it develops strictly from the ovary.

- Papayas and Grapes are true fruits (botanically classified as berries) because they develop exclusively from a fertilised ovary.

Additional Facts:

- ✓ In most plants, by the time the fruit develops from the ovary, other floral parts degenerate and fall off. However, in false fruits, the thalamus (or other floral parts) contributes significantly to fruit formation and becomes the fleshy, edible portion.
- ✓ Examples of false fruit:
 - Apple: The edible fleshy part is entirely the swollen thalamus. The actual ovary forms the tough, papery core containing the seeds.
 - Strawberry: The red, fleshy part we eat is the swollen receptacle (thalamus). The tiny “seeds” on the outside are actually individual true fruits called achenes.
 - Pear: Similar to the apple, the fleshy part is the thalamus.
 - Cashew: The fleshy “cashew apple” develops from the pedicel/thalamus.
 - Mulberry: The entire inflorescence (cluster of flowers) swells to form the fruit, making it a composite false fruit.

Q 24. Ans. B

Explanation:

- Li-Fi (Light Fidelity) transmits data using the visible light spectrum (via flickering LED bulbs) rather than radio waves.
- **Additional facts:**
 - ✓ **Bandwidth:** The visible light spectrum is roughly 10,000 times larger than the entire radio frequency spectrum, offering massively higher data capacities.
 - ✓ **Security:** Because visible light cannot penetrate solid walls, the signal is confined to the room. This restricts the range but makes it virtually immune to external hacking.
 - ✓ **Key Applications:** Highly useful in environments where radio frequencies cause dangerous interference, such as aircraft cabins, hospital operating rooms, and deep underwater operations.

Q 25. Ans. B**Explanation:**

- The human tongue is a freely movable muscular organ. Its upper dorsal surface is covered with distinct, small macroscopic projections. These structural projections are anatomically classified as **papillae**. Some of these papillae house the taste buds required for gustation.
- **Microvilli** are microscopic cellular membrane protrusions. They exist at the cellular level to increase surface area for diffusion. They are predominantly found on the epithelial cells lining the **small intestine**.
- **Rugae** are macroscopic ridges or folds of mucosal tissue. They provide structural elasticity to allow an organ to expand its internal volume. They are predominantly found in the internal lining of the empty human **stomach**.
- **Villi** are macroscopic finger-like tissue projections. They serve to vastly increase the internal absorptive surface area of the digestive tract. They are exclusively located in the mucosal lining of the **small intestine**.

Q 26. Ans. D**Explanation:**

- “CAR” stands for **Chimeric Antigen Receptor**. These are synthetic, laboratory-engineered receptors designed to bind to specific proteins (antigens) on the surface of cancer cells.
- **Additional facts:**
 - ✓ Mechanism (Living Drug): It is a form of immunotherapy. A patient's own immune cells (T-cells) are extracted from their blood, genetically altered in a lab to grow CARs on their surface, and then multiplied and infused back into the patient to hunt and destroy cancer cells.
 - ✓ Currently, it is most highly effective and approved for liquid tumours/blood cancers, such as B-cell acute lymphoblastic leukaemia (ALL) and lymphomas, typically targeting the CD19 protein.
 - ✓ In 2023, the Central Drugs Standard Control Organisation (CDSCO) approved NexCAR19, India's first indigenously developed CAR T-cell therapy (developed by ImmunoACT, IIT Bombay, and Tata Memorial Hospital). This indigenous development reduced the therapy's cost from roughly ₹3-4 crore (abroad) to around ₹40 lakh in India.

Q 27. Ans. B**Explanation:**

- **Statement 1 is incorrect:** The gallbladder does not possess the biological cellular machinery to synthesise bile. Bile juice is exclusively synthesised and secreted by hepatic cells located within the liver. The human gallbladder is structurally a thin muscular sac. Its primary physiological function is to temporarily store the bile delivered from the liver via the hepatic ducts. The gallbladder also actively absorbs water from the stored bile to highly concentrate it. This concentrated bile is eventually released into the cystic duct. The actual emulsification of large fat globules into smaller micelles is performed by the bile salts contained within this fluid.
- **Statement 2 is correct:** The duodenum is the C-shaped proximal portion of the small intestine. It receives the combined digestive secretions of the liver and the pancreas. These distinct secretions travel together through a shared anatomical vessel. This vessel is known as the hepato-pancreatic duct. The final release of these fluids into the duodenum is strictly regulated by a circular muscle. This specific muscular valve is called the sphincter of Oddi.

Q 28. Ans. C**Explanation:**

- The gastric glands located in the mucosal lining of the stomach contain three major types of secretory cells. **Parietal cells** are one of these major cell types. They are alternatively known as oxyntic cells. These specific cells are exclusively responsible for the active secretion of Hydrochloric Acid (HCl) into the stomach lumen. They also secrete a substance called intrinsic factor. Intrinsic factor is a vital glycoprotein essential for the intestinal absorption of vitamin B12.
- The secreted HCl creates a highly acidic environment within the gastric cavity. The optimal pH of this environment is approximately 1.8. This highly acidic medium is necessary for the initiation of protein digestion. It triggers the chemical conversion of the inactive proenzyme pepsinogen into the active proteolytic enzyme pepsin. Pepsin then actively digests dietary proteins into smaller peptides like proteoses and peptones. The HCl also provides a rudimentary immunological defence by destroying many pathogenic microbes ingested with food. Other surrounding gastric cells secrete

a thick layer of mucus and bicarbonates. This alkaline mucosal layer protects the delicate stomach epithelium from physical excoriation and chemical burning by the highly concentrated acid.

- Peptic cells are another distinct cell type located deep within the gastric glands. They are alternatively called chief cells. These cells secrete the inactive precursor enzyme called pepsinogen. They do not possess the biological machinery to secrete acid.
- Mucus neck cells are present in the upper region of the gastric glands. They continuously secrete a thick mucus. This mucus forms the primary physical protective barrier over the gastric epithelium.
- Goblet cells are specialised glandular epithelial cells. They are predominantly found in the mucosal lining of the intestinal tract rather than the gastric glands of the stomach. They secrete mucus to lubricate the intestinal lumen. This continuous lubrication facilitates the smooth passage of digested food and faecal matter.

Q 29. Ans. A

Explanation:

- **Statement 1 is correct:** During fertilisation, a sperm cell contributes only its nuclear DNA; its mitochondria (located in the tail to power swimming) are either left outside the egg or actively destroyed by the egg after entry. Consequently, a child inherits 100% of their mitochondrial genome from the mother.
- **Statement 2 is correct:** Mitochondria are responsible for cellular respiration and energy (ATP) generation. Mutations in mtDNA lead to an energy crisis within cells, which preferentially devastates highly active tissues.
- **Statement 3 is incorrect:** There is currently no cure for mitochondrial diseases in living adults. Mitochondrial Replacement Therapy (MRT) is an in vitro fertilisation (IVF) technique designed solely to prevent the transmission of the disease from an affected mother to her future embryo.
- **Additional facts:**
 - ✓ Three-Parent Baby: Offspring born via MRT possess genetic material from three individuals: the nuclear DNA from the biological mother and father (~99.8%), and the healthy mitochondrial DNA from a third-party female donor (~0.2%).
 - ✓ Key techniques of MRT: The two primary methods approved in certain jurisdictions (like the UK) are Pronuclear Transfer (PNT)

and Maternal Spindle Transfer (MST). Both involve swapping the nuclear material into a donor egg possessing healthy mitochondria.

- ✓ Mitochondrial genome structure: Unlike human nuclear DNA, which is arranged in linear chromosomes, mitochondrial DNA is circular, closely resembling bacterial DNA. This supports the endosymbiotic theory (that mitochondria evolved from ancient bacteria engulfed by early eukaryotic cells).

Q 30. Ans. C

Explanation:

- **Statement 1 is correct:** Endocrine glands are anatomically defined by their complete lack of secretory ducts. They are universally referred to as ductless glands. These specialised glands synthesise biochemical messengers called hormones. They release these hormones directly into the interstitial fluid surrounding the glandular tissue. These hormones subsequently diffuse into the systemic venous circulation for transport to distant target organs.
- **Statement 2 is correct:** Exocrine glands possess a distinct structural network of ducts. They utilise these ducts to transport their secretions to internal epithelial surfaces or the external body surface. Their secretions include non-hormonal products such as mucus, earwax, sebum, saliva, milk, and digestive enzymes. Examples include salivary glands, sweat glands, and pancreatic acinar cells.

Q 31. Ans. B

Explanation:

- **Statement 1 is correct:** ESCs, found in the inner cell mass of an early embryo (3-5 days post-fertilisation), are pluripotent. They are the master blank slates capable of generating all tissues of the body.
- **Statement 2 is incorrect:** Induced Pluripotent Stem Cells (iPSCs) are not naturally extracted from birth tissues. They are artificially created in a laboratory by introducing specific genes into adult, specialised somatic cells (like skin or blood cells), forcing them to revert backwards into an embryonic, "stem-cell-like" state.
- **Statement 3 is correct:** Adult Stem Cells are undifferentiated cells found among differentiated ones in a tissue. They are limited in their potential, acting primarily as a repair system for their native tissue (e.g., neural stem cells only make brain cells).

● **Additional facts:**

- ✓ Potency Hierarchy:
 - Totipotent: Can form the entire organism plus extra-embryonic tissues (placenta). Example: The zygote.
 - Pluripotent: Can form all cells of the adult body, but not the placenta. Example: ESCs, iPSCs.
 - Multipotent: Can form multiple, but strictly related, cell types. Example: Hematopoietic (blood) stem cells in bone marrow.
- ✓ Nobel Prize: The discovery that mature cells can be reprogrammed to become pluripotent (iPSCs) won the 2012 Nobel Prize in Medicine (awarded to Shinya Yamanaka and John Gurdon), as it bypassed the massive ethical debate surrounding the destruction of human embryos to harvest ESCs.

Q 32. Ans. C

Explanation:

- The defining characteristic of a phenotype (and by extension, the phenome) is that it is heavily influenced by the environment. Two organisms with the exact same genome (like identical twins or cloned plants) can have different phenomes if exposed to different diets, climates, or stress factors.
- **Additional facts:**
 - ✓ Genome: The complete set of DNA instructions found in a cell (The blueprint).
 - ✓ Transcriptome: The complete set of RNA transcripts produced by the genome under specific circumstances (The active blueprints being read).
 - ✓ Proteome: The entire set of proteins expressed by a genome, cell, tissue, or organism (The actual building blocks and machines).
 - ✓ Phenome: The total sum of all observable traits resulting from the proteins interacting with the environment (The final, visible structure).

Q 33. Ans. B

Explanation:

- The pituitary gland is a small endocrine gland. It is anatomically located at the base of the human brain. It sits securely within a protective bony cavity. This specific cavity is named the **sella turcica**. The sella turcica is a structural depression. This depression is a physical feature of the sphenoid bone of the cranium. The gland is physically attached to the overlying hypothalamus by a slender stalk.

- **Foramen magnum:** It is a large anatomical opening. It is located at the base of the occipital bone. It serves as the primary exit point for the central nervous system from the skull. The spinal cord passes through this opening to connect with the brain stem.
- **Cribriform plate:** It is a porous bony structure. It is a distinct anatomical component of the ethmoid bone. It contains numerous small perforations. These perforations transmit olfactory nerves from the nasal cavity directly into the cranial vault.
- **Mandibular fossa:** It is a distinct bony depression. It is located on the inferior surface of the temporal bone. It articulates directly with the mandible. This specific articulation forms the movable temporomandibular joint of the jaw.

Q 34. Ans. B

Explanation:

- Cloves are the desiccated, pre-anthesis (unopened) flower buds of *Syzygium aromaticum*, an evergreen tree belonging to the Myrtaceae family. The buds are harvested manually when the calyx transitions from green to a distinct crimson-pink, and are subsequently sun-dried. The primary bioactive compound responsible for their distinct aroma and analgesic properties is eugenol.
- **Additional facts:**
 - ✓ Spices can be derived from almost any part of the plant.
 - ✓ Classification of major spices categorised by the specific plant organ harvested:
 - Floral Parts (Buds and Stigmas)
 - ◇ Clove (*Syzygium aromaticum*): Unopened flower buds.
 - ◇ Saffron (*Crocus sativus*): The dried, tripartite stigmas and styles (female reproductive structures) of the flower.
 - ◇ Capers (*Capparis spinosa*): The unburst, pickled flower buds of a Mediterranean shrub.
 - Arils and Seeds
 - ◇ Nutmeg and Mace (*Myristica fragrans*): This is a unique species that yields two distinct spices. Nutmeg is the actual dried seed (kernel), while Mace is the fleshy, lacy aril that surrounds that seed.
 - ◇ Coriander (*Coriandrum sativum*): The dried schizocarpic fruits (commonly referred to as seeds).

- ◇ Cumin (*Cuminum cyminum*): The dried seed-like fruit (achene).
- ◇ Fenugreek (*Trigonella foenum-graecum*): The cuboid-shaped dried seeds of a leguminous plant.
- Bark (Phloem Tissue)
 - ◇ True Cinnamon (*Cinnamomum verum*): The dried inner bark (primarily phloem tissue) is native to Sri Lanka.
 - ◇ Cassia / Chinese Cinnamon (*Cinnamomum cassia*): A thicker, more pungent bark derived from a related species, often sold commercially as standard cinnamon.
- Fruits and Berries
 - ◇ Black Pepper (*Piper nigrum*): The dried, unripe drupes (berries) of a climbing vine. (White pepper is the same seed with the fleshy pericarp removed).
 - ◇ Cardamom (*Elettaria cardamomum*): The dried, unripe capsular fruits contain multiple seeds.
 - ◇ Vanilla (*Vanilla planifolia*): The cured, elongated seed pods (capsules) of a climbing orchid.
 - ◇ Chilli Peppers (*Capsicum* spp.): The hollow, fleshy berries of the plant are rich in the alkaloid capsaicin.
- Modified Underground Stems (Rhizomes)
 - ◇ Ginger (*Zingiber officinale*): The thick, creeping underground rhizome.
 - ◇ Turmeric (*Curcuma longa*): The boiled, dried, and powdered rhizome, characterised by the presence of the compound curcumin.

Q 35. Ans. B

Explanation:

- A biosensor detects a specific biological molecule (like a biochemical marker) and converts that detection into a measurable electrical signal.
- **Additional facts:**
 - ✓ A standard glucometer used by diabetics. The test strip contains an enzyme (the biological element) that reacts with glucose in the blood. The machine (the detector) converts that chemical reaction into a digital number on the screen.
 - ✓ Pregnancy tests: A simple, non-electronic biosensor that uses antibodies to detect the presence of the hCG hormone (a biochemical marker) in urine.

Q 36. Ans. B

Explanation:

- Iodine is an essential dietary micronutrient. It is strictly required for the normal synthesis of thyroid hormones. These hormones include tetraiodothyronine (T₄) and triiodothyronine (T₃). A chronic deficiency of dietary iodine directly reduces the internal production of these hormones. This specific clinical condition is termed **hypothyroidism**. The thyroid gland attempts to physiologically compensate for this reduced hormonal output. It undergoes a compensatory physical enlargement. This visibly enlarged thyroid gland is medically referred to as a **goitre**.
- **Hypoglycemia:** It is a physiological condition characterised by abnormally low blood glucose levels. It is typically associated with the overproduction of insulin by the islets of Langerhans. The pancreas does not physically enlarge due to a lack of dietary iodine.
- **Hyperthyroidism:** It is characterised by an excessive, rather than deficient, production of thyroid hormones. It is often caused by thyroid cancer or autoimmune nodules. A specific form called exophthalmic goitre causes the physical protrusion of the eyeballs. It is distinctly not caused by a nutritional iodine deficiency.
- **Hypocalcemia:** It is a state of abnormally low calcium levels in the systemic blood. It can lead to rapid, sustained muscle spasms known as tetany. This condition is directly linked to the undersecretion of parathyroid hormone (PTH) by the parathyroid glands.

Q 37. Ans. A

Explanation:

- **Option A is correct:** Osmosis is a specific type of diffusion where water (the solvent) moves passively down its concentration or water potential gradient across a semipermeable cell membrane.
- **Option B is incorrect:** This describes active transport. While roots do use active transport to absorb certain mineral ions, the absorption of water via osmosis is a strictly passive process requiring no ATP.
- **Option C is incorrect:** This describes the movement of solutes (minerals), not the movement of the solvent (water). Furthermore, osmosis specifically involves a semi-permeable or selectively permeable membrane.

- **Option D is incorrect:** This describes imbibition, a special type of diffusion where water is adsorbed by solids/colloids (like a dry seed swelling up), not the osmotic transfer across root cell membranes.
- **Additional Facts:**
 - ✓ **Root Hairs:** Water is primarily absorbed by root hairs, which are delicate, unicellular elongations of the root epidermis that vastly increase the surface area for absorption.
 - ✓ **Water Potential:** Pure water has the highest possible water potential, defined as zero. When solutes (minerals) are added to the root cytoplasm, its water potential becomes negative, creating the gradient that pulls soil water inward.
 - ✓ **Apoplast vs. Symplast:** Once inside the root hair, water travels to the xylem via two pathways: the apoplast pathway (through the non-living cell walls and intercellular spaces) and the symplast pathway (through the living cytoplasm and plasmodesmata).
 - ✓ **Casparian Strip:** Located in the endodermis of the root, this waxy, suberized band blocks the apoplast pathway. It forces all water and minerals to pass through the cell membrane (symplast), allowing the plant to filter and control what enters the xylem.

Q 38. Ans. B

Explanation:

- Blood plasma is a straw-coloured and viscous fluid. It constitutes nearly 55% of the total blood volume. Water is the primary component of plasma. It accounts for approximately 90 to 92% of the plasma. Proteins contribute about 6 to 8% of the total plasma volume.
- Albumins are the most abundant proteins found in the plasma. These proteins are primarily responsible for maintaining the osmotic balance of the blood. They prevent the excessive leakage of fluid into the surrounding body tissues.
- Fibrinogen is an essential protein produced by the liver and found in the plasma. It is a critical factor in the process of blood coagulation. Plasma that does not contain these clotting factors is referred to as serum.
- Electrolytes are inorganic minerals present in the plasma in small quantities. These include sodium, calcium, magnesium, bicarbonate, and chloride ions. They are vital for maintaining the proper pH and osmotic pressure of the body.

- Haemoglobin is not a component of the blood plasma. It is a complex iron-containing protein pigment. It is located exclusively within the red blood cells. Haemoglobin is responsible for the binding and transport of respiratory gases. It gives the red blood cells their characteristic colour. Its absence from the plasma prevents it from being filtered out by the kidneys.

Q 39. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Carnivorous (insectivorous) plants are typically endemic to acidic bogs and wetlands. In these environments, waterlogging prevents rapid bacterial decomposition of organic matter, leading to a severe depletion of bioavailable nitrogen. To synthesise nucleic acids and proteins, these plants evolved complex anatomical traps to digest insects and extract their nitrogenous compounds.
- **Pair 2 is correctly matched:** Pneumatophores are specialised “breathing roots” developed by halophytes (salt-tolerant plants) like mangroves. They inhabit coastal swamps where the muddy, waterlogged soil creates a strictly anaerobic environment, leading to a severe oxygen deficiency for root respiration. Pneumatophores grow upward (against gravity) out of the mud and water to absorb oxygen directly from the atmosphere via microscopic pores called lenticels.
- **Pair 3 is incorrectly matched:** Leguminous plants form symbiotic root nodules in association with Rhizobium bacteria to adapt to soils deficient in nitrogen, not potassium. The bacteria possess the nitrogenase enzyme, enabling them to fix inert atmospheric nitrogen into usable ammonia for the plant, while the plant provides carbohydrates in return.
- **Additional facts:**
 - ✓ **Plants Adapted to Nitrogen-Deficient Soils (Carnivorous Plants)**
 - **Pitcher Plant:** Leaves are modified into a deep cavity filled with digestive enzymes to drown and dissolve prey.
 - **Venus Flytrap:** They have active snap-traps triggered by sensitive mechanoreceptor hairs on the leaf surface.
 - **Sundew:** Utilises “flypaper” traps covered in sticky, glandular tentacles that ensnare and digest insects.
 - **Bladderwort:** Aquatic or sub-aquatic plants featuring tiny, vacuum-driven bladder traps to capture micro-invertebrates.

- Butterwort: Employs sticky, resinous leaves to trap small gnats and midges.
- ✓ Plants Adapted to Oxygen-Deficient Anaerobic Soils (Mangroves/Halophytes)
 - Red Mangrove: Highly recognisable by their arching “stilt roots” that provide structural support and aid in aeration.
 - Black Mangrove: Characterised by dense clusters of pencil-like pneumatophores protruding vertically from the anoxic mud.
 - Mangrove Apple: Produces thick, conical pneumatophores for survival in high-salinity, low-oxygen tidal zones.
 - Sundari Tree: The dominant mangrove species of the Sundarbans, notable for its extensive pneumatophore network.
- ✓ Plants Adapted to Nitrogen-Deficient Soils via Symbiosis (Legumes)
 - Soybean
 - Chickpea
 - Alfalfa
 - Peanut
 - Lentil

Q 40. Ans. A

Explanation:

- **Statement 1 is correct:** The human ovary produces two primary groups of steroid hormones. These are estrogen and progesterone. Estrogen is actively synthesised and secreted mainly by the growing ovarian follicles. This specific hormone is biologically responsible for the stimulation of growth in female secondary sex organs. It also drives the development of growing ovarian follicles. Furthermore, estrogen regulates the appearance of female secondary sexual characteristics. Examples of these characteristics include a high-pitched voice and initial mammary gland development.
- **Statement 2 is incorrect:** The corpus luteum is a temporary endocrine structure. It is formed from the remnants of the ruptured ovarian follicle immediately following ovulation. This specific structure is responsible for the predominant secretion of progesterone. However, progesterone does not initiate or stimulate the appearance of secondary sexual characteristics. Progesterone is fundamentally responsible for supporting and maintaining pregnancy. It achieves this by maintaining the endometrial lining of the uterus. Progesterone also acts directly on the mature mammary glands. It stimulates the formation of sac-like alveoli necessary for internal milk storage.

Q 41. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Machine Learning (ML) is the fundamental subset of AI where systems are fed data to find patterns and make decisions autonomously, moving away from rigid, rule-based coding.
- **Pair 2 is incorrectly matched:** Natural Language Processing (NLP) is not hardware. It is the branch of AI concerned with giving computers the ability to understand text and spoken words in much the same way human beings can (e.g., translation, sentiment analysis). The description provided refers to hardware accelerators like TPUs (Tensor Processing Units) or NPUs.
- **Pair 3 is correctly matched:** Large Language Models (LLMs) are a highly advanced subset of ML and NLP. They use massive deep neural networks to recognise, summarise, translate, predict, and generate text.
- **Pair 4 is correctly matched:** A Generative Pre-trained Transformer (GPT) relies on the Transformer architecture (introduced by Google in 2017). Unlike older models (RNNs) that processed text one word at a time, Transformers use “self-attention” to look at an entire **sentence at once**, **understanding context and allowing for massive parallel processing**.
- **Additional facts:**
 - ✓ The “Attention” Mechanism: The core breakthrough of the Transformer architecture. It allows the AI to weigh the importance of different words in a sentence, regardless of their distance from each other, giving models like ChatGPT human-like contextual understanding.
 - ✓ Parameters: In LLMs, “parameters” act like the artificial synapses in a brain. They are the variables the model adjusts during training to minimise errors. GPT-3, for example, has 175 billion parameters.
 - ✓ Generative AI vs. Discriminative AI: Discriminative AI classifies existing data (e.g., identifying if an image is a cat or a dog). Generative AI (like GPT or Midjourney) creates entirely new data, text, or images that did not previously exist.

Q 42. Ans. D

Explanation:

- **Statement 1 is correct:** The pancreas is an elongated organ. It is anatomically situated between the

limbs of the C-shaped duodenum. It is classified as a heterocrine or composite gland. This means it contains both exocrine and endocrine tissue.

- **Statement 2 is correct:** The exocrine part of the pancreas is composed of groups of cells called acini. These cells secrete an alkaline pancreatic juice. This juice is transported through the pancreatic duct to the duodenum. It contains essential digestive enzymes. These enzymes include trypsinogen, chymotrypsinogen, and steapsin. These enzymes facilitate the breakdown of proteins, fats, and carbohydrates in the small intestine.
- **Statement 3 is correct:** The endocrine portion of the pancreas consists of small clusters of cells called the Islets of Langerhans. There are approximately 1 to 2 million islets in a normal human pancreas. These islets represent only 1 to 2% of the total pancreatic tissue. They contain two primary types of hormone-secreting cells. The alpha-cells are responsible for secreting the hormone glucagon. Glucagon acts to increase blood glucose levels. The beta-cells are responsible for secreting the hormone insulin. Insulin acts to decrease blood glucose levels. Both hormones are critical for maintaining the homeostatic balance of blood sugar.

Q 43. Ans. A

Explanation:

- **Statement 1 is correct:** Quantum Key Distribution (QKD) is a major application of quantum communication. It uses the principles of quantum mechanics to transmit cryptographic keys securely. Any physical attempt to intercept the key alters its quantum state and immediately alerts the communicating parties.
- **Statement 2 is correct:** Classical computers struggle to model the complex quantum states of large molecules. Quantum computers can simulate these complex molecular and chemical interactions simultaneously. This drastically reduces the computational time required for discovering new pharmaceuticals and advanced materials.
- **Statement 3 is correct:** Quantum sensors exploit quantum states to achieve unprecedented measurement sensitivity. Technologies like quantum interferometers and advanced atomic clocks provide extreme precision. They enable accurate positioning and navigation in environments where traditional GPS signals are jammed or unavailable.
- **Statement 4 is incorrect:** Autonomous smart

contracts operate on decentralised, peer-to-peer blockchain networks. This is a defining characteristic of Web 3.0 technology, not quantum technology. In fact, advanced quantum computing poses a theoretical threat to the current RSA encryption algorithms that secure modern blockchain networks.

- **Statement 5 is correct:** Quantum computers utilise the principle of quantum superposition to process massive numbers of variables and possibilities at the exact same time. This makes them highly effective for solving complex optimisation problems, such as streamlining global logistical routes or managing financial risk portfolios.

Additional facts:

- ✓ **National Quantum Mission (NQM):** Approved by the Union Cabinet and implemented by the Department of Science & Technology (DST) with a budget of over ₹6,000 crores.
- ✓ **Core Domains:** The mission focuses on four primary verticals: Quantum Computing, Quantum Communication, Quantum Sensing & Metrology, and Quantum Materials & Devices.
- ✓ **Quantum Superposition:** The fundamental principle that enables a qubit (quantum bit) to represent both 0 and 1 simultaneously, unlike classical bits, which are strictly binary.
- ✓ **Quantum Entanglement:** A phenomenon where multiple qubits become intrinsically linked. The state of one entangled qubit instantly dictates the state of another, regardless of physical distance, enabling rapid internal data transfer.

Q 44. Ans. D

Explanation:

- Individuals with the **AB positive (AB+)** blood group are classified as universal acceptors. This classification arises because their red blood cells possess all three primary surface antigens: Antigen A, Antigen B, and the Rh antigen. Consequently, their blood plasma does not contain any naturally occurring antibodies against these antigens. Since their immune system recognises all three antigens as "self," they can safely receive a transfusion of any blood type without triggering an immunological reaction or erythrocyte agglutination.
- **O negative (O-)** group: It is classified as the universal donor. Red blood cells of this group completely lack A, B, and Rh surface antigens. This absence allows

O- blood to be transfused into any recipient with minimal risk of an immediate immune response. However, O- individuals can only receive blood from other O- donors because their plasma contains antibodies against all other antigens.

- **AB negative (AB-) group:** While it lacks anti-A and anti-B antibodies, it does not possess the Rh antigen. If an AB- individual receives Rh-positive blood, their immune system will produce anti-Rh antibodies. Therefore, they are not true universal acceptors as they cannot safely receive Rh-positive blood types.
- **O positive (O+) group:** This is the most common blood type. It contains the Rh antigen but possesses both anti-A and anti-B antibodies in the plasma. These antibodies will aggressively attack any transfused red blood cells that carry A or B antigens. Thus, O+ individuals can only receive O+ or O- blood.

Q 45. Ans. B

Explanation:

- The Petiole is the distinct stalk that attaches the leaf blade (lamina) to the stem node. Beyond structural support, its primary physiological function is to hold the leaf blade outward to maximise light interception for photosynthesis. Furthermore, its flexible nature allows the leaf to flutter in the wind, which facilitates a cooling effect and maintains a steady supply of fresh air (carbon dioxide) to the leaf surface.
- Pedicel is the stalk of an individual flower within an inflorescence or a cluster.
- Peduncle is the main supporting stalk of an entire inflorescence (a cluster of flowers) or the stem of a solitary, terminal flower.
- Stipules are the small, paired, often leaf-like appendages found at the very base of the petiole where it meets the stem. They primarily serve to protect the young leaf bud before it unfolds.
- **Additional facts:**
 - ✓ **Petiolate Leaves:** Leaves that possess a distinct petiole.
 - Examples: Mango (*Mangifera indica*), Peepal (*Ficus religiosa*), Hibiscus (*Hibiscus rosa-sinensis*), and most dicotyledonous plants.
 - ✓ **Sessile Leaves:** Leaves that completely lack a petiole, meaning the leaf blade is attached directly to the stem.
 - Examples: Wheat (*Triticum aestivum*),

Rice (*Oryza sativa*), Safflower (*Carthamus tinctorius*), and many monocotyledonous grasses.

- ✓ **Petiole Modifications (Phyllodes)** In certain xerophytic (desert-adapted) plants, the true leaf blades are highly reduced or fall off early to minimise water loss. In these cases, the petiole itself is modified by flattening and turning green to take over the function of photosynthesis. This modified petiole is termed a phyllode.
 - Example: Australian Acacia (*Acacia auriculiformis*).

Q 46. Ans. D

Explanation:

- **Statement 1 is correct:** A reflex action is an automatic response to a peripheral nervous stimulation. It occurs without conscious effort or thought. The entire neural pathway involved in this action is termed a reflex arc. This pathway ensures an extremely rapid response to potentially harmful stimuli.
- **Statement 2 is correct:** The sensory neuron is also known as the afferent neuron. It receives signals from sensory receptors like the skin or eyes. It conducts these nerve impulses toward the central nervous system. In most spinal reflexes, these impulses enter the dorsal root of the spinal cord.
- **Statement 3 is correct:** The motor neuron is also known as the efferent neuron. It receives processed signals from the central nervous system. It carries these instructions away from the spinal cord. The impulses are delivered to an effector organ such as a muscle or a gland. This triggers a physical action like muscle contraction.
- A complete reflex arc typically involves at least one afferent neuron and one efferent neuron arranged in a series. Many reflex arcs also include a relay neuron. The relay neuron is situated entirely within the spinal cord. It serves as a biological bridge between the sensory and motor neurons. It allows for the rapid integration of signals without involving the brain. The receptor organ first detects the stimulus. The effector organ finally executes the resulting motor response.

Q 47. Ans. A

Explanation:

- Plants, like all living organisms, require a continuous supply of energy to survive. They perform cellular

respiration 24 hours a day, absorbing oxygen and releasing carbon dioxide to generate ATP. During the day, this carbon dioxide release is masked by a much higher rate of photosynthesis, which consumes carbon dioxide. At night, in the absence of sunlight, photosynthesis stops completely, making the steady release of carbon dioxide from respiration dominant.

● **Additional facts:**

- ✓ **Cellular respiration location:** While photosynthesis occurs exclusively in chloroplasts, found mainly in green plant parts, cellular respiration occurs in the mitochondria of all living plant cells, including roots, stems and leaves.
- ✓ **Net gaseous exchange:**
 - Daytime: Net release of oxygen and net intake of carbon dioxide.
 - Nighttime: Net release of carbon dioxide and net intake of oxygen.
- ✓ **Stomatal closure:** In most plants, stomata close at night to reduce water loss through transpiration, but they do not seal completely. Sufficient gaseous exchange continues through partially closed stomata, lenticels in stems, and pneumatophores in mangrove roots to sustain respiration.

Q 48. Ans. C

Explanation:

- **Statement 1 is correct:** Although milk contains a wide variety of nutrients necessary for growth, it is notably deficient in Iron. This is a critical point in pediatric nutrition as prolonged exclusive milk feeding can lead to iron-deficiency anaemia.
- **Statement 2 is incorrect:** Milk is a very poor source of Vitamin C. Any small amounts naturally present are often further reduced or destroyed during the pasteurisation process due to heat sensitivity.
- **Statement 3 is correct:** Milk protein consists of approximately 80% Casein and 20% Whey. Both are considered “complete” proteins because they contain all nine essential amino acids required by the human body.
- **Additional facts:**
 - ✓ **Mineral Balance:** While deficient in iron, milk is an excellent source of Calcium and Phosphorus, which are present in a ratio that promotes optimal bone absorption.
 - ✓ **Vitamin Fortification:** Because milk is a staple, many countries (including India via FSSAI

standards) practice voluntary or mandatory fortification of milk with Vitamin A and Vitamin D to combat widespread deficiencies.

- ✓ **Carbohydrates:** The primary sugar in milk is Lactose. Individuals lacking the enzyme lactase cannot digest this sugar, leading to a clinical condition known as lactose intolerance.
- ✓ **Riboflavin (B2):** Milk is a major source of Vitamin B2. The slight yellowish tint sometimes seen in milk (especially in cow milk) is partly due to the presence of riboflavin and carotene.
- ✓ **B12 Source:** For vegetarians, milk and dairy products are among the few natural dietary sources of Vitamin B12, which is essential for nerve function and RBC formation.

Q 49. Ans. C

Explanation:

- **Statement 1 is correct:** In terms of vascular tissue evolution, Gymnosperms are more primitive. Their xylem is primarily composed of tracheids and lacks true vessel elements (with the rare exception of the order Gnetales). Similarly, their phloem contains sieve cells and albuminous cells, but lacks the specialised sieve tubes and companion cells that efficiently transport nutrients in Angiosperms.
- **Statement 2 is correct:** In Angiosperms, double fertilisation occurs: one male gamete fuses with the egg (forming the zygote), and the other fuses with the diploid secondary nucleus to form a triploid (3n) endosperm, which nourishes the embryo. In contrast, Gymnosperms do not exhibit double fertilisation. Their endosperm represents the vegetative tissue of the female gametophyte itself; thus, it is haploid (n) and forms before fertilisation takes place.
- **Additional Facts:**
 - ✓ **Naked Seed:**
 - Gymnosperms (from Greek gymnos = naked, sperma = seed) produce ovules that are not enclosed by any ovary wall. Consequently, they remain exposed both before and after fertilisation. Because there is no ovary, Gymnosperms do not produce true fruits.
 - Angiosperms possess flowers where the ovules are enclosed within an ovary. Post-fertilisation, the ovules mature into seeds, and the surrounding ovary matures into a fruit.

- ✓ Examples of Gymnosperms:
 - Pine: A monoecious plant (both male and female cones on the same tree) known for highly efficient wind pollination. Its pollen grains have distinctive “wings.”
 - Cycas: A dioecious plant (separate male and female trees) often planted as an ornamental palm-like tree. It harbours specialised coralloid roots that contain symbiotic cyanobacteria (Anabaena) for nitrogen fixation.
 - Maidenhair Tree: Often referred to as a “living fossil” because it is the sole surviving species of its entire evolutionary order, having remained virtually unchanged for over 200 million years.
 - Sequoia sempervirens: The Giant Redwood tree, a gymnosperm, holds the record for being one of the tallest tree species in the world.
- ✓ Pollination Mechanisms
 - Gymnosperms rely almost exclusively on anemophily (wind pollination), which requires the production of massive quantities of lightweight pollen.
 - Angiosperms have co-evolved extensively with animals, utilising zoophily (pollination by insects, birds, and bats) in addition to wind and water, which is a major factor in their current evolutionary dominance.

Q 50. Ans. C

Explanation:

- The cerebrum is the largest and most prominent part of the human brain. It is the primary component of the forebrain. This region serves as the main thinking centre of the central nervous system. It possesses specialised areas for receiving sensory impulses from various receptors. These areas interpret information related to hearing, smell, and sight. The cerebrum is also responsible for complex cognitive processes such as reasoning, logic, and memory. It facilitates the conscious control of voluntary muscle movements.
- The cerebellum is a smaller but vital part of the hindbrain. It is located at the posterior base of the brain. This structure is essential for the precision of voluntary actions. It coordinates muscle groups to allow for complex tasks like riding a bicycle or walking in a straight line. The cerebellum

continuously monitors and maintains the physical posture of the body. It regulates the equilibrium and balance of the individual during movement.

- Both structures work in harmony within the central nervous system. While the cerebrum decides to perform an action, the cerebellum ensures that the action is performed smoothly and accurately. The forebrain, midbrain, and hindbrain together form the integrated biological system that controls and coordinates every human activity.

Q 51. Ans. D

Explanation:

- **Vivipary** is a highly specialised reproductive strategy observed in many halophytes, particularly mangroves (e.g., Rhizophora). In the harsh, highly saline, and anaerobic intertidal mudflats, a dormant seed would rapidly succumb to salt toxicity or rot before germinating. To bypass this, the seed germinates in situ on the mother plant, drawing nutrients until it develops into a robust seedling (propagule). It then drops like a weighted dart into the mud, ready to establish roots immediately.
- **Apomixis** is a form of asexual reproduction in plants where viable seeds are produced without the fusion of gametes (fertilisation), mimicking sexual reproduction.
- **Cleistogamy** refers to an anatomical adaptation where flowers remain permanently closed, strictly enforcing self-pollination (autogamy) and preventing cross-pollination.
- **Parthenocarpy** is the development of a fruit without prior fertilisation of the ovule, naturally resulting in a seedless fruit (e.g., commercial bananas).
- **Additional facts:**
 - ✓ Halophytes are plants that have evolved to survive and complete their life cycles in environments with high soil salinity, such as coastal swamps, estuaries, and saline semi-deserts.
 - ✓ Their primary evolutionary adaptations include:
 - Morphological adaptations
 - ◇ Pneumatophores: Negatively geotropic “breathing roots” equipped with lenticels to facilitate gaseous exchange in anoxic, waterlogged soils.
 - ◇ Stilt and Prop Roots: Extensive aerial root systems provide mechanical stabilisation against tidal forces and shifting mud.

- ◊ Succulence: Many coastal halophytes (e.g., *Salicornia*) develop thick, fleshy leaves and stems. This allows them to store large volumes of water to internally dilute accumulated salt concentrations.
- Physiological adaptations
 - ◊ Salt Exclusion: Specialised root endodermis actively filters and prevents the massive influx of sodium and chloride ions into the xylem stream.
 - ◊ Salt Excretion: Species like the Black Mangrove (*Avicennia*) possess epidermal salt glands on their leaves that actively secrete excess absorbed salt, which crystallises on the leaf surface.
 - ◊ Osmotic Adjustment: Halophytes accumulate compatible organic solutes (like proline or glycine betaine) in their cellular vacuoles to lower their internal water potential, preventing dehydration by maintaining the osmotic gradient necessary to draw water from the saline soil.
- ✓ All viruses have a protein coat called a capsid that protects the nucleic acid. Some viruses (like HIV or SARS-CoV-2) have an additional lipid envelope derived from the host's cell membrane.
- ✓ Antibiotics target cellular structures (like cell walls) or metabolic pathways specific to bacteria. Since viruses are acellular and use the host's machinery, antibiotics have no effect on them.
- ✓ Smaller than viruses, consisting only of free RNA without a protein coat (mostly cause plant diseases).
- ✓ Infectious agents composed entirely of abnormally folded proteins are responsible for diseases like "Mad Cow Disease."
- ✓ A specific group of RNA viruses (like HIV) that use an enzyme called reverse transcriptase to convert their RNA into DNA once inside the host.

Q 53. Ans. C

Explanation:

- **Statement 1 is correct:** Chlorophyll molecules primarily absorb wavelengths in the blue (around 430-450 nm) and red (around 640-680 nm) regions of the visible light spectrum. These are the wavelengths that drive photosynthesis most efficiently.
- **Statement 2 is incorrect:** This is a classic conceptual trap. Plants appear green because chlorophyll reflects or transmits green light, not because it absorbs it. It is the least absorbed wavelength.
- **Statement 3 is correct:** An absorption spectrum plots the wavelengths of light absorbed, while an action spectrum plots the rate of photosynthesis against wavelength. The two graphs closely mirror each other, proving that chlorophyll 'a' is the chief pigment responsible for photosynthesis.
- **Additional Facts:**
 - ✓ **Photosynthetically Active Radiation (PAR):** This is the portion of the light spectrum utilised by plants for photosynthesis, ranging strictly from 400 nm to 700 nm.
 - ✓ **Accessory pigments:** While chlorophyll a is the primary pigment, others like chlorophyll b, xanthophylls (yellow), and carotenoids (yellow to yellow-orange) act as accessory pigments. They absorb a broader range of light and transfer the energy to chlorophyll a.
 - ✓ **Photo-oxidation protection:** Beyond trapping light, carotenoids serve a crucial secondary

Q 52. Ans. B

Explanation:

- **Statement 1 is correct:** Viruses are not cells. They lack the protoplasm, cell membrane, and organelles (like ribosomes or mitochondria) that define cellular life.
- **Statement 2 is correct:** In the environment, viruses are metabolically inactive "virions." They only show signs of life (replication) once they enter a compatible living host cell.
- **Statement 3 is incorrect:** A virus contains either DNA or RNA as its genetic material, but never both in the same particle. This is a fundamental distinction from cellular organisms, which **contain both**.
- **Statement 4 is correct:** Bacteriophages (literally "bacteria eaters") are viruses that specifically target bacterial cells for infection and reproduction.
- **Additional facts:**
 - ✓ Viruses are often described as being on the "threshold of life." They are considered non-living because they can be crystallised and lack independent metabolism, yet living because they possess genetic material and can mutate.

function; they protect chlorophyll a from photo-oxidation (destruction by excessive light).

- ✓ **Engelmann's experiment:** T.W. Engelmann first described the action spectrum of photosynthesis by splitting light using a prism and observing where aerobic bacteria congregated around a green alga (*Cladophora*). The bacteria predominantly accumulated in the blue and red light regions.

Q 54. Ans. B

Explanation:

- **Option A is incorrect:** This statement describes fertilisation (specifically syngamy), which is a different subsequent event that occurs only after successful pollination and pollen tube growth.
- **Option B is correct:** In biological terms, pollination is defined as the mechanism of transferring pollen grains (shed from the anther) to the receptive surface of the stigma. It is a prerequisite step that makes fertilisation possible.
- **Option C is incorrect:** This describes pollen germination and pollen tube growth, a post-pollination event that bridges the gap between pollination and fertilisation.
- **Option D is incorrect:** This describes dehiscence (the bursting open of the anther to release pollen), which is only the starting point, not the actual transfer to a female reproductive organ.
- **Additional facts:**
 - ✓ **Kinds of Pollination:**
 - Autogamy: Transfer of pollen to the stigma of the same flower (self-pollination).
 - Geitonogamy: Transfer of pollen to the stigma of another flower on the same plant (functionally cross-pollination but genetically similar to autogamy).
 - Xenogamy: Transfer of pollen to the stigma of a different plant (true cross-pollination, bringing genetic variation).
- **Agents of Pollination:** Plants use both abiotic (wind - anemophily, water - hydrophily) and biotic (insects - entomophily, birds - ornithophily, bats - chiropterophily) agents to achieve this transfer.

Q 55. Ans. A

Explanation:

- **Statement 1 is correct:** The human heart is a four-chambered muscular organ. It consists of two relatively small upper chambers called atria. It also

contains two larger lower chambers called ventricles. A thick muscular wall called the interatrial septum separates the right and left atria. Similarly, an interventricular septum separates the right and left ventricles. The opening between the right atrium and the right ventricle is guarded by the tricuspid valve. This valve consists of three muscular flaps or cusps. It prevents the backflow of blood into the atrium during ventricular contraction.

- **Statement 2 is incorrect:** The human cardiovascular system performs double circulation. This process consists of two distinct pathways: pulmonary circulation and systemic circulation. Pulmonary circulation begins in the right ventricle. Deoxygenated blood is pumped into the pulmonary artery toward the lungs. Oxygenated blood then returns to the left atrium via the pulmonary veins. Systemic circulation begins in the left ventricle. Oxygenated blood is pumped into the aorta to reach all body tissues. This pathway provides nutrients and oxygen to the cells. It simultaneously carries carbon dioxide and other harmful substances away for elimination. Deoxygenated blood from the tissues is eventually collected by the vena cava and emptied into the right atrium.
- The bicuspid valve is another critical internal structure. It is alternatively known as the mitral valve. It guards the opening between the left atrium and the left ventricle. This valve consists of two muscular flaps. Semilunar valves are located at the base of the pulmonary artery and the aorta. These valves ensure that blood flows in only one direction. They prevent blood from flowing back into the ventricles from the major arteries. The entire heart is enclosed in a double-walled membranous bag called the pericardium. This structure contains pericardial fluid to reduce friction during heartbeats.

Q 56. Ans. B

Explanation:

- **Hemodynamics** refers to the study of the physical forces governing blood circulation.
- Blood pressure is the lateral pressure exerted by blood against the walls of the arteries. A **sphygmomanometer** is the instrument used to measure this pressure. The measurement is recorded in units of millimetres of mercury. A healthy adult typically has a blood pressure of 120/80. The number 120 represents the systolic or pumping pressure. The number 80 represents the diastolic or resting pressure.

- A **stethoscope** is an acoustic medical device. It is used to listen to the internal sounds of the human body. In cardiology, it detects the “lubb” and “dubb” sounds caused by the closure of heart valves.
- An **electrocardiograph** is a machine used to obtain an electrocardiogram. It records the electrical activity of the heart over a period of time. It detects the electrical impulses that trigger heart contractions.
- A **spirometer** is an apparatus for measuring the volume of air inspired and expired by the lungs. It is used to assess pulmonary function. It helps in determining clinical respiratory volumes and capacities.

Q 57. Ans. A

Explanation:

- **Statement 1 is correct:** For a long time, scientists incorrectly believed that the oxygen released during photosynthesis came from carbon dioxide. However, microbiologist Cornelius van Niel proposed, and later isotope experiments confirmed, that the oxygen gas released by green plants originates exclusively from the light-driven splitting of water, a process known as photolysis.
- **Statement 2 is incorrect:** While cyanobacteria (blue-green algae) carry out oxygenic photosynthesis and release oxygen, purple sulphur bacteria and green sulphur bacteria perform anoxygenic photosynthesis. Instead of using water as the electron donor, they use hydrogen sulphide, and therefore release sulphur or sulphate compounds rather than oxygen gas.
- **Additional facts:**
 - ✓ The splitting of water (photolysis) is physically associated with the Photosystem II (PS II) complex on the inner side of the thylakoid membrane. This crucial process requires specific micronutrients, most notably Manganese (Mn) and Chlorine (Cl).
 - ✓ Ruben and Kamen Experiment: Definitive proof came from isotope tracer experiments using the heavy oxygen isotope O-18. When plants were supplied with water containing O-18, the oxygen gas released also contained O-18. However, when carbon dioxide containing O-18 was supplied, the released oxygen did not contain the isotope. This proved that oxygen evolves from water, not carbon dioxide.
 - ✓ The shift from anoxygenic to oxygenic photosynthesis by early cyanobacteria is responsible for the “Great Oxidation

Event” roughly 2.4 billion years ago, which permanently altered Earth’s atmosphere and allowed aerobic life to evolve.

Q 58. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Zoospores are microscopic, motile structures that use flagella to swim, serving as the primary method of asexual reproduction in many algae (e.g., Chlamydomonas).
- **Pair 2 is correctly matched:** Gemmae are specialised, multicellular asexual buds that develop in small receptacles called gemma cups on the thallus of liverworts (e.g., Marchantia). They detach to grow into new individuals.
- **Pair 3 is incorrectly matched:** The protonema is a creeping, green, branched filamentous stage of the gametophyte in Mosses (Bryophytes), not Pteridophytes.
- **Pair 4 is incorrectly matched:** The prothallus is the inconspicuous, multicellular, haploid gametophyte in Pteridophytes (ferns). It is not a diploid sporophyte, nor is it associated with mosses.
- **Additional Facts:**
 - ✓ Isogamy vs. Oogamy: In lower plants like algae, sexual reproduction ranges from isogamous (fusion of morphologically identical gametes, e.g., Spirogyra) to oogamous (fusion of a large non-motile female egg and a small motile male sperm, e.g., Volvox).
 - ✓ The “Amphibians” of the Plant Kingdom: Bryophytes and Pteridophytes are often given this title because, while they live on land, they are strictly dependent on a continuous film of water for their ciliated male gametes (antherozoids) to swim to the female sex organ (archegonium) for fertilisation.
 - ✓ Alternation of Generations: Plant life cycles alternate between a haploid (n) gamete-producing phase and a diploid (2n) spore-producing phase.
 - ✓ In Bryophytes (mosses/liverworts), the main plant body is the haploid gametophyte.
 - ✓ In Pteridophytes (ferns), the evolutionary shift occurs, making the diploid sporophyte the dominant, independent main plant body.

Q 59. Ans. A

Explanation:

- **Statement 1 is correct:** Arteries are the primary blood vessels responsible for carrying blood away

from the heart. They transport this blood to various organs and tissues throughout the body. The heart actively pumps blood directly into these arterial vessels. This mechanical pumping action places the emerging blood under significantly high pressure. Consequently, arteries must possess thick, elastic muscular walls. These highly resilient walls allow the vessels to safely stretch and withstand the internal hydrostatic pressure without rupturing.

- **Statement 2 is incorrect:** Veins are the specific blood vessels that collect blood from different peripheral organs. They return this collected blood back to the heart. The blood flowing within the venous system is no longer under high pressure. Therefore, veins structurally possess comparatively thin walls. Veins are uniquely equipped with distinct internal valves. These internal valves function as physical, one-way gates. They guarantee that the slow-moving, low-pressure blood flows exclusively in the direction of the heart. This specific structural adaptation actively prevents the backward flow and pooling of blood within the lower extremities due to gravity.

Q 60. Ans. B

Explanation:

- While standard generative AI (like early ChatGPT) waits for a human prompt, generates an answer, and stops, Agentic AI acts as an autonomous agent. If given a broad goal (e.g., "Research the top 5 solar panel companies and email me a summary report"), it will independently break the task into steps, browse the web, compile the data, and send the email without needing a human to prompt it at every single step.
- **Additional facts:**
 - ✓ **Tool Integration:** The core power of Agentic AI is its ability to use external digital tools autonomously. It can be granted access to APIs to read emails, execute code, manage calendars, or conduct financial transactions on a user's behalf.
 - ✓ **Evolution of AI:** It represents the shift from Conversational AI (AI as a chatbot or advisor) to Action-Oriented AI (AI as a digital worker or proxy).
 - ✓ **Prominent Examples:** Experimental open-source projects like AutoGPT and BabyAGI were early demonstrations of agentic architecture.
 - ✓ **India's first agentic AI assistant, Kruti,** has been developed by Krutrim.

Q 61. Ans. D

Explanation:

- **Statement 1 is correct:** The most visible function of a stem is to act as the main structural axis of the plant, spreading out branches and leaves so they can receive optimal sunlight for photosynthesis.
- **Statement 2 is correct:** The stem acts as a two-way biological highway. It transports water and dissolved minerals from the roots upwards (via xylem) and distributes the food manufactured in the leaves to other plant parts (via phloem).
- **Statement 3 is incorrect:** While mostly aerial, stems can be modified to grow completely underground for the purpose of storing food and surviving harsh conditions. The most common everyday example of this is the potato.
- **Additional Facts:**
 - ✓ **Vegetative Propagation:** Because stems possess nodes and buds, modified stems like potatoes (tubers), onions (bulbs), and ginger (rhizomes) can be used to naturally reproduce new plants without seeds.
 - ✓ **Photosynthetic Stems:** In desert plants like the cactus, the leaves are reduced to spines to prevent water loss. In these cases, the stem takes over the function of making food and becomes thick, green, and fleshy.
 - ✓ **Support and Protection:** Weak stems can modify themselves into thread-like tendrils (as seen in grapevines or gourds) to climb for support, or into hard, woody thorns (as in citrus plants or bougainvillea) to protect the plant from grazing animals.

Q 62. Ans. A

Explanation:

- **Statement 1 is correct:** Tidal volume is the specific volume of air moved into or out of the lungs during a normal breathing cycle. This physiological process occurs without any conscious effort or forceful muscle contraction. This volume is approximately 500 millilitres per breath in a healthy human adult. A healthy man can inspire or expire approximately 6000 to 8000 millilitres of air per minute.
- **Statement 2 is incorrect:** Vital capacity is the maximum volume of air a person can breathe in after a forced expiration. It can also be measured as the maximum volume of air a person can breathe out after a forced inspiration. This functional capacity includes the inspiratory reserve volume, the tidal volume, and the expiratory reserve volume.

- **Residual volume** is the distinct volume of air that constantly remains in the lungs even after a maximal forced expiration. This specific residual air ensures that the microscopic alveoli do not physically collapse during the breathing cycle.

Q 63. Ans. A

Explanation:

- **Option A is correct:** 4D printing is a direct technological extension of 3D printing. The “fourth dimension” involved in this process is time. The physical object is printed using specialised smart materials. These materials possess the ability to autonomously transform their shape, size, or mechanical properties post-production. This dynamic transformation is triggered by specific external environmental stimuli, such as heat, light, moisture, or magnetic fields.
- **Option B is incorrect:** This describes advanced multi-material bioprinting. While this is a cutting-edge medical technology used for tissue engineering, it does not define the time-dependent transformation inherent to the concept of 4D printing.
- **Option C is incorrect:** This refers to predictive artificial intelligence modelling and digital twin technology in additive manufacturing. It is a software-based quality control mechanism rather than a physical property of the printed material itself.
- **Option D is incorrect:** This describes AR/VR (Augmented/Virtual Reality) assisted additive manufacturing. This technology enhances the digital design and visualisation phase. It does not alter the physical capabilities of the final printed object after the manufacturing process is complete.
- **Additional Facts:**
 - ✓ **Smart materials:** The foundation of 4D printing relies heavily on materials like Shape Memory Alloys (SMAs), Shape Memory Polymers (SMPs), and responsive hydrogels.
 - ✓ **Biomedical applications:** 4D printing is heavily researched for targeted drug delivery. A 4D-printed microscopic capsule can travel through the bloodstream and unfold to release medication only when it detects the specific temperature or pH of a tumour site.
 - ✓ **Aerospace and defence:** The technology allows for self-assembling structures. Large solar panels or antennas could be printed

compactly on Earth, launched into space, and programmed to autonomously unfold when exposed to solar radiation.

- ✓ **Infrastructure:** The technology enables the creation of adaptive piping systems. These pipes can dynamically expand or contract their diameter in response to the changing flow rate or temperature of the water inside them.

Q 64. Ans. B

Explanation:

- **Low Earth Orbit (LEO)** is strictly defined as an orbit located under an altitude of 2000 km. It is primarily utilised for Earth observation, military reconnaissance, and the International Space Station (ISS) due to its close proximity to the surface.
- **Sun-Synchronous Orbit (SSO)** is a highly specific type of near-polar orbit typically located between 600 to 800 km. It ensures the satellite passes over any specific geographic point at the exact same local solar time. It is a part of Low Earth Orbit (LEO).
- **Medium Earth Orbit (MEO)** occupies the vast spatial region from 2,000 to 36,000 kilometres. It is the primary orbital domain utilised by global navigation satellite systems (GNSS) such as the American GPS or the European Galileo.
- **Geostationary Orbit (GEO)** is precisely fixed at an altitude of exactly 35,786 km directly over the Earth’s equator. A satellite placed here perfectly matches the Earth’s rotational speed. It appears completely stationary in the sky to a ground observer.
- **Additional facts:**
 - ✓ **Geosynchronous vs Geostationary:** A geosynchronous orbit has a period equal to Earth’s rotation (24 hours) but can be inclined. A geostationary orbit is a specific type of geosynchronous orbit that has zero inclination, meaning it sits precisely over the equator.
 - ✓ **Orbital Velocity:** Satellites in LEO must travel much faster (approximately 7.8 km/s) to counteract the stronger gravitational pull. Satellites in GEO travel significantly slower (approximately 3 km/s).
 - ✓ **Communication Delay:** Because GEO is located at such a high altitude, signals sent to these satellites experience a noticeable latency (delay) of about 250 milliseconds, which makes them less ideal for high-speed, real-time gaming or automated trading compared to LEO constellations like Starlink.

Q 65. Ans. C

Explanation:

- **Statement A is correct:** Haemoglobin is a complex protein pigment found in red blood cells, and its core contains iron. A deficiency in this iron leads to anaemia, a common nutritional disorder.
- **Statement B is correct:** Erythrocytes (RBCs) act as the body's primary transport vehicles for respiratory gases, picking up oxygen in the lungs and distributing it to all the living cells for cellular respiration.
- **Statement C is incorrect:** Haemoglobin actually has a significantly higher affinity for carbon monoxide (CO) than for oxygen, about 200 times more. If a person breathes in smoke or vehicular exhaust containing carbon monoxide, the CO strongly binds to haemoglobin, preventing oxygen from attaching. This leads to carbon monoxide poisoning and suffocation, even if oxygen is present in the air.
- **Statement D is correct:** Unlike most other cells in the human body, mature red blood cells do not have a nucleus. This unique structural adaptation creates a biconcave shape and frees up maximum internal volume to pack in as much haemoglobin as possible.

Q 66. Ans. C

Explanation:

- **Quantum Determinism** is not a fundamental principle of quantum technology. Classical physics operates on strict determinism, where exact outcomes can be predicted if initial conditions are known. Quantum mechanics, conversely, is inherently probabilistic. It is physically impossible to predict the precise outcome of a single quantum event with absolute certainty.
- **Quantum Superposition** is a core defining principle. It enables a quantum system, such as a qubit, to exist in multiple distinct states at the exact same time. This state collapses into a single classical value only upon direct measurement.
- **Quantum Entanglement** is a proven foundational phenomenon. It inextricably links two or more subatomic particles. The physical measurement of one particle instantaneously dictates the exact state of its entangled partner, regardless of the spatial distance separating them.
- **Quantum Interference** is a vital operational mechanism in quantum computing. It dictates how quantum wave functions interact. Quantum

algorithms specifically use this principle to constructively amplify the probability of reaching a correct computational answer while destructively cancelling out incorrect pathways.

● Additional Facts:

- ✓ **Heisenberg's Uncertainty Principle:** This principle is the exact reason quantum determinism does not exist. Formulated by Werner Heisenberg, it states that it is impossible to simultaneously measure both the exact position and exact momentum of a subatomic particle.
- ✓ **Algorithmic Application:** Shor's Algorithm relies heavily on quantum interference and superposition to factor large prime numbers exponentially faster than classical computers, posing a theoretical threat to modern RSA cryptography.

Q 67. Ans. C

Explanation:

- **Statement 1 is correct:** Macronutrients (carbohydrates, proteins, and fats) are needed in gram quantities. They serve as the primary fuel for the body and provide the raw materials for tissue growth and repair.
- **Statement 2 is correct:** While micronutrients (vitamins and minerals) are vital for health, they do not have any caloric value. Instead, they act as essential components for enzyme activity and physiological regulation.
- **Additional facts:**
 - ✓ **Energy density:** Fats provide approximately 9 kcal/g, while carbohydrates and proteins provide about 4 kcal/g. Micronutrients provide 0 kcal/g.
 - ✓ **Storage:** The body can store large amounts of macronutrients (as glycogen or adipose tissue), whereas many micronutrients (especially water-soluble vitamins) cannot be stored and must be consumed regularly.
 - ✓ **Metabolic role:** Macronutrients are "oxidised" to produce ATP; micronutrients act as "co-factors" or "co-enzymes" that enable these oxidative reactions to occur.

Q 68. Ans. B

Explanation:

- **Statement 1 is incorrect:** The upward movement of water and minerals through the xylem is a passive process. It relies almost entirely on physical

forces, primarily the suction pull generated by transpiration (evaporation from leaves) and the cohesive/adhesive properties of water. It does not require the expenditure of ATP by the plant.

- **Statement 2 is correct:** The translocation of food (primarily sucrose) in the phloem is an active process. While the actual mass flow of sap is driven by turgor pressure, the initial loading of sugars into the phloem at the source (leaves) and unloading at the sink (roots/fruits) requires metabolic energy in the form of ATP.

- **Additional Facts:**

- ✓ **Cellular Nature:** The main conducting elements of Xylem (tracheids and vessels) are structurally dead and heavily lignified at maturity. In contrast, the conducting elements of Phloem (sieve tube elements) are living cells, though they lack a nucleus and depend on adjacent companion cells.
- ✓ **Bidirectional vs. Unidirectional:** Xylem transport is strictly unidirectional (roots to leaves). Phloem transport is bidirectional (or multidirectional) because the “source” (where food is made) and “sink” (where food is needed) can change depending on the season or plant’s needs.
- ✓ **Form of Transport:** Plants synthesise food as glucose, transport it primarily as sucrose (a non-reducing, relatively inert disaccharide), and store it as starch.
- ✓ **Girdling Experiment:** A classic experiment used to identify tissue function. If a ring of bark (which includes the phloem) is removed from a tree trunk, the tissue above the ring swells due to accumulated food, proving that phloem is responsible for downward food translocation.

Q 69. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The Small Satellite Launch Vehicle (SSLV) is a three-stage launch vehicle. All three of its stages are powered by solid propulsion. It is designed specifically to launch small commercial satellites. It has the capability to place a payload of up to 500 kg into a 500 km Low Earth Orbit (LEO).
- **Pair 2 is correctly matched:** The Polar Satellite Launch Vehicle (PSLV) is the highly reliable workhorse of ISRO. It is a four-stage rocket. It uses alternating solid and liquid propulsion stages. It is

primarily designed to deliver payloads of up to 1,750 kg into Sun-Synchronous Polar Orbits (SSPO).

- **Pair 3 is correctly matched:** The Launch Vehicle Mark-3 (LVM3) is ISRO’s heaviest and most powerful operational rocket. It was formerly known as GSLV Mk-III. It utilises two solid strap-on motors, a liquid core stage, and a high-thrust cryogenic upper stage. It can successfully place a 4,000 kg payload into a Geosynchronous Transfer Orbit (GTO).
- **Pair 4 is incorrectly matched:** The Geosynchronous Satellite Launch Vehicle (GSLV Mk-II) is a three-stage vehicle. It does not possess the thrust required to carry 10,000 kg. Its maximum payload capacity is approximately 6,000 kg to Low Earth Orbit. It can carry up to 2,500 kg to a Geosynchronous Transfer Orbit.

- **Additional Facts:**

- ✓ **Gaganyaan Mission:** ISRO’s LVM3 has been re-engineered as the Human Rated Launch Vehicle (HRLV). It is the designated rocket for India’s upcoming crewed spaceflight program, Gaganyaan.
- ✓ **Cryogenic Technology:** The upper stage of the LVM3 uses the indigenous CE-20 cryogenic engine. It uses liquid oxygen and liquid hydrogen stored at extremely low temperatures to generate massive thrust.
- ✓ **SSLV Commercial Advantage:** The SSLV was developed strictly for the “launch-on-demand” market. It requires minimal launch infrastructure. It can be assembled and launched by a small team within just 72 hours, drastically reducing commercial launch costs.
- ✓ **Historic PSLV Missions:** The PSLV was the launch vehicle used for India’s most historic early space exploration missions. These include Chandrayaan-1 (2008) and the Mars Orbiter Mission (Mangalyaan) in 2013.

Q 70. Ans. C

Explanation:

- **Platelets**, also scientifically known as **thrombocytes**, are specialised cell fragments produced in the bone marrow. Their primary function is to prevent excessive blood loss from the body during an injury. Whenever a blood vessel is damaged, platelets rush to the site and clump together to form a physical plug. They also release specific biochemical factors that trigger the complex blood coagulation (clotting) cascade. A significant reduction in the

platelet count, often seen in diseases like Dengue, can lead to severe clotting disorders and life-threatening internal or external haemorrhages.

- The transport of oxygen is the exclusive function of Red Blood Cells (erythrocytes), which contain the protein haemoglobin.
- The production of antibodies and the fight against foreign pathogens is the primary responsibility of White Blood Cells (leukocytes), specifically the lymphocytes.
- The maintenance of osmotic pressure and pH balance is primarily managed by blood plasma and its dissolved proteins, such as albumins, along with the regulatory functions of the kidneys.

Q 71. Ans. C

Explanation:

- A **Mesh network topology** connects nodes directly and non-hierarchically to as many other nodes as possible. This decentralised structure provides unparalleled redundancy. If one specific node or connection fails, the network automatically reroutes the data through alternative active pathways to ensure continuous communication.
- A **Star network topology** connects all peripheral nodes to a single, centralised hub or switch. While easy to manage, it contains a critical single point of failure. If the central hub fails, the entire network collapses entirely.
- A **Ring network topology** connects nodes sequentially in a continuous closed loop. Data typically travels in only one direction. A single point of failure, such as one broken cable or inactive node, can disrupt the entire communication chain.
- A **Bus network topology** connects all individual nodes to a single central cable, known as the backbone. If this main backbone cable breaks or experiences severe interference, the entire network goes down.
- **Additional Facts:**
 - ✓ **IoT & Smart Homes:** Mesh networking is the foundational technology for modern smart home communication protocols like Zigbee and Z-Wave. It allows low-power devices to bounce signals off each other to reach a central router.
 - ✓ **Military & Disaster Relief:** The Armed Forces and disaster response teams utilise Tactical Mobile Ad Hoc Networks (MANETs). These are self-forming, self-healing mesh networks designed to operate in hostile environments without relying on pre-existing, fixed cellular towers.

- ✓ **Full vs. Partial Mesh:** In a Full Mesh, every single node physically connects to every other node, offering maximum security but high cost. In a Partial Mesh, only the most critical routing nodes are fully interconnected to optimise infrastructure costs.

Q 72. Ans. C

Explanation:

- **Biomass Mission** was not launched by NASA. It is an Earth observation mission developed by the European Space Agency (ESA). It holds the distinction of being the first satellite to carry a P-band Synthetic Aperture Radar (SAR). Its primary objective is to map the world's forests to provide comprehensive measurements of forest biomass and globally stored carbon.
- **ESCAPADE** (Escape and Plasma Acceleration and Dynamics Explorers) mission is a NASA initiative. It consists of twin spacecraft designed to investigate Mars. It specifically studies how the solar wind interacts with the Martian magnetic environment and how this interaction drives the planet's atmospheric escape.
- **Lucy Mission** is a NASA space probe launched in 2021. It is the very first space mission dedicated to exploring the Jupiter Trojan asteroids. These asteroids are primitive remnants from the early solar system that orbit the Sun in two distinct "swarms" leading and following Jupiter.
- **Orbiting Carbon Observatory (OCO)** is a dedicated NASA Earth observation mission. Its primary objective is to collect precise, global measurements of atmospheric carbon dioxide levels to better understand the Earth's carbon cycle and climate change dynamics.
- **Additional Facts:**
 - ✓ **LUPEX Mission:** The Lunar Polar Exploration (LUPEX) mission, also known as Chandrayaan-5, is a joint venture between India's ISRO and Japan's JAXA. ISRO will primarily build the lander, while JAXA will provide the rover to study volatile materials at the lunar South Pole.
 - ✓ **Tianwen-2 Probe:** This is an upcoming mission by the China National Space Administration (CNSA). Its ambitious objective is to collect physical samples from the near-Earth asteroid 2016HO3 and further explore the main-belt comet 311P.
 - ✓ **Solar Orbiter:** This is a collaborative mission

between the ESA and NASA. It is designed to take unprecedented close-up images of the Sun and observe the uncharted Solar Polar Regions to understand the solar magnetic field.

Q 73. Ans. C

Explanation:

- **GLONASS** is a global navigation satellite system developed and operated by Russia. It was fully operationalised in 1995. It serves as a direct, global alternative to the American GPS system.
- **Galileo** is the global navigation satellite system created by the European Union. It is operated by the European Space Agency (ESA). Unlike GPS or GLONASS, it is strictly under civilian control rather than military control.
- **BeiDou** is the global satellite navigation system owned and operated by China. It recently completed its third-generation constellation (BDS-3). It provides continuous positioning, navigation, and timing services worldwide.
- **NavIC** (Navigation with Indian Constellation) is an independent regional navigation satellite system. It is designed and operated completely by India. It provides an accurate position information service to users in India and the region extending up to 1500 km from its boundary.
- **Additional Facts:**
 - ✓ Global Positioning System (GPS): The most widely used system globally, owned by the United States government and operated by the United States Space Force.
 - ✓ IRNSS Structure: NavIC was formerly known as the Indian Regional Navigation Satellite System (IRNSS). Its constellation consists of seven active satellites. Three are located in Geostationary Orbit (GEO), and four are in inclined Geosynchronous Orbit (GSO).
 - ✓ Global vs. Regional: Systems like GPS, GLONASS, Galileo, and BeiDou are Global Navigation Satellite Systems (GNSS) requiring dozens of satellites placed in Medium Earth Orbit (MEO). NavIC is distinctly a Regional system.
 - ✓ Dual Frequency: NavIC provides two distinct services. The Standard Positioning Service (SPS) is provided to all civilian users. The Restricted Service (RS) is an encrypted service provided exclusively to authorised military and security agencies.

Q 74. Ans. B

Explanation:

- **Option A is incorrect:** This describes a solid or hybrid rocket motor. Solid propellants are chemically stable and stored at ambient temperatures. They do not require the complex, super-cooled storage infrastructure of a cryogenic stage.
- **Option B is correct:** A cryogenic engine uses liquid oxygen (LOX) and liquid hydrogen (LH₂) as propellants. These gases are liquefied by cooling them to extremely low temperatures. Liquid hydrogen is stored at -253°C. Liquid oxygen is stored at -183°C. When mixed and ignited in the combustion chamber, they expand rapidly. This chemical reaction generates a massive, highly efficient thrust.
- **Option C is incorrect:** This describes an Ion Thruster. Ion engines use electrical power to accelerate heavy ions like xenon. They provide incredibly high fuel efficiency but extremely low physical thrust. They are suitable only for deep-space manoeuvring, not lifting payloads against Earth's gravity.
- **Option D is incorrect:** This describes a monopropellant engine. These engines use a single chemical, such as hydrazine. It decomposes instantly upon hitting a heated catalyst bed to produce gas. They are primarily used for small satellite attitude control, not main-stage heavy lifting.
- **Additional Facts:**
 - ✓ Specific Impulse (Isp): Cryogenic engines offer the highest specific impulse among all chemical rockets. For every kilogram of propellant burned, they produce significantly more thrust than conventional solid or Earth-storable liquid engines.
 - ✓ ISRO's Indigenous Milestone: The Indian Space Research Organisation (ISRO) successfully developed its own cryogenic upper stage engine, known as the CE-20.
 - ✓ Launch Vehicle Mark-3 (LVM3): The CE-20 engine specifically powers the crucial upper stage of the LVM3. This is India's heaviest operational rocket, responsible for launching the Chandrayaan-3 mission and the upcoming Gaganyaan human spaceflight mission.

Q 75. Ans. C

Explanation:

- **Lymph** is a clear, colourless fluid that plays a crucial role in the human immune and transport systems.

When blood passes through capillaries, some water and dissolved substances leak out into the tissue spaces to bathe the cells. This is called tissue fluid or interstitial fluid. The lymphatic system, a parallel network of microscopic vessels, collects this excess tissue fluid. Once inside these lymphatic vessels, the fluid is officially termed “lymph.” It is eventually drained back into the major veins (subclavian veins) near the collarbone, successfully returning the fluid to the bloodstream. Lymph also carries infection-fighting white blood cells (lymphocytes) and absorbs digested fats from the intestine.

- **Plasma:** It is the straw-coloured, liquid matrix of the blood itself, flowing within the main blood vessels and carrying red blood cells, white blood cells, and platelets.
- **Serum:** It is specifically defined as blood plasma from which the clotting proteins (like fibrinogen) have been completely removed. I
- **Chyle:** It is a type of lymph, and it is specifically a milky fluid consisting of fat droplets and lymph that drains solely from the small intestine during digestion, not the general interstitial fluid collected from all body tissues.

Q 76. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Project Kuiper is a massive satellite internet initiative. It is being heavily funded and developed directly by Amazon.
- **Pair 2 is correctly matched:** Starlink is currently the most prominent and recognisable satellite internet constellation globally. It is exclusively developed, launched, and operated by SpaceX.
- **Pair 3 is incorrectly matched:** OneWeb is not associated with Blue Origin. It is operated by the French satellite operator Eutelsat. It currently maintains the world's second-largest satellite constellation fleet after SpaceX.
- **Pair 4 is incorrectly matched:** The Qianfan constellation is a major planned space communications project. It is specifically a Chinese low-Earth orbit (LEO) satellite internet megaconstellation.
- **Additional Facts:**
 - ✓ LEO Megaconstellations: Modern satellite internet relies heavily on Low Earth Orbit (LEO). Placing satellites under 2,000 km altitude drastically reduces the signal latency that plagued older Geostationary (GEO) internet satellites.
 - ✓ Bridging the Digital Divide: The ultimate

socio-economic objective of these competing global projects is universal connectivity. They are uniquely capable of providing reliable, high-speed broadband to difficult mountainous or rural terrains where laying underground optical fibre is economically unviable.

Q 77. Ans. D

Explanation:

- **Trachoma** is a highly contagious bacterial infection. It primarily affects the human eyes. It causes the inner surface of the eyelids to become severely inflamed and scarred. This scarring eventually turns the eyelashes inward. The inward-turned eyelashes physically scratch the cornea. This continuous friction leads to irreversible blindness.
- Lungs are primarily affected by lower respiratory tract infections. Common bacterial examples include Tuberculosis and Pneumonia. They are not the biological target organ for Trachoma.
- Liver is primarily affected by hepatotropic viral infections. Common examples include Hepatitis B and Hepatitis C. Trachoma bacteria do not infect hepatic tissue.
- Kidneys are primarily affected by conditions like glomerulonephritis or ascending urinary tract infections. They are completely unrelated to the clinical pathogenesis of Trachoma.
- **Additional Facts:**
 - ✓ At the 78th World Health Assembly in Geneva, India was officially awarded the Certificate of Elimination of Trachoma as a Public Health Problem by the World Health Organisation (WHO). This recognition follows the elimination of the infectious eye disease, making India the third country in the WHO South-East Asia Region to achieve this milestone.
 - ✓ Causative Agent: The disease is caused by a specific obligate intracellular bacterium named *Chlamydia trachomatis*.
 - ✓ Vector Transmission: It spreads rapidly through direct personal contact with infected eye or nose secretions. It is also mechanically transmitted by eye-seeking flies that breed in unhygienic conditions.
 - ✓ SAFE Strategy: The WHO heavily recommends the “SAFE” strategy for global Trachoma elimination. This comprehensive acronym stands for Surgery (for late-stage disease), Antibiotics (to clear infection), Facial cleanliness, and Environmental improvement.

Q 78. Ans. C

Explanation:

- **Rod cells:** The retina contains two major types of photoreceptor cells. Rods contain a purplish-red protein called rhodopsin (visual purple), which contains a derivative of Vitamin A. They are highly sensitive to dim light and are responsible for twilight or night vision (scotopic vision). They do not play a role in colour differentiation.
- **Cone cells:** Cones are responsible for daylight vision (photopic vision) and colour vision. There are three specific types of cone cells in the human eye that respond to red, green, and blue light. The sensation of different colours is produced by various combinations of these cones and their photopigments.
- **Blind spot:** The optic nerve leaves the eye and the retinal blood vessels enter it at a specific point slightly above the posterior pole of the eyeball. Photoreceptor cells (both rods and cones) are completely absent in this specific region. Therefore, no image is formed here, earning it the name “blind spot.”
- **Fovea centralis:** At the posterior pole of the eye, lateral to the blind spot, there is a yellowish pigmented spot called the macula lutea with a central pit called the fovea. The fovea is a thinned-out portion of the retina where only cone cells are densely packed. It is the exact point where visual acuity (resolution of the image) is the greatest.

Q 79. Ans. C

Explanation:

- **Pair A is correctly matched:** Hepatitis A is a highly contagious liver infection caused by the Hepatitis A virus (HAV). Unlike the other strains, it is primarily transmitted via the faecal-oral route, usually through the consumption of contaminated drinking water or unhygienic food. It typically causes an acute, self-limiting illness rather than a chronic disease.
- **Pair B is correctly matched:** Hepatitis B is unique among the common hepatitis viruses because it possesses a double-stranded DNA genome (all others are RNA viruses). It is transmitted through exposure to infectious blood, semen, and other body fluids, including transmission from an infected mother to her infant at birth.
- **Pair C is incorrectly matched:** There is currently no vaccine available to prevent Hepatitis C. It is a blood-borne RNA virus most commonly

transmitted through sharing unsterilized needles or unsafe medical practices. However, while there is no vaccine, modern Direct-Acting Antiviral (DAA) medicines can completely cure over 95% of Hepatitis C infections.

- **Pair D is correctly matched:** Hepatitis D (the Delta virus) is an incomplete, defective RNA virus. It cannot replicate or cause infection inside the human body on its own. It strictly requires the helper function of the Hepatitis B virus (HBV) surface antigen to multiply in liver cells. Therefore, Hepatitis D only occurs as a co-infection or super-infection in individuals who already have Hepatitis B.

Additional Facts:

- ✓ **Universal Immunisation Programme (UIP):** The Government of India provides the Hepatitis B vaccine free of cost under the UIP. The first dose is crucially administered to infants within 24 hours of birth to prevent vertical (mother-to-child) transmission.
- ✓ **Chronic Liver Disease:** Hepatitis B and C are the most dangerous strains globally because they frequently lead to chronic (lifelong) infections. If left untreated, they are the leading causes of liver cirrhosis (severe scarring) and hepatocellular carcinoma (liver cancer).
- ✓ **Nobel Prize 2020:** The Nobel Prize in Physiology or Medicine in 2020 was jointly awarded to Harvey J. Alter, Michael Houghton, and Charles M. Rice for their groundbreaking discovery of the Hepatitis C virus, which led to the development of life-saving antiviral drugs.

Q 80. Ans. B

Explanation:

- **Statement A is incorrect:** While eukaryotes (plants, animals, fungi) have a true membrane-bound nucleus, prokaryotes (bacteria and archaea) are complete, independently living cells that entirely lack a defined nucleus. Their genetic material is naked and lies free in the cytoplasm.
- **Statement B is correct:** Unicellular organisms are capable of independent existence and performing the essential functions of life. Therefore, anything less than a complete structure of a cell does not ensure independent living. This is exactly why entities like viruses are considered to be on the borderline of living and non-living; they lack a complete cellular structure and their own metabolic machinery, forcing them to rely entirely on a host cell's machinery to replicate.

- **Statement C is incorrect:** While mitochondria are the “powerhouses” of eukaryotic cells, prokaryotic cells do not possess membrane-bound organelles like mitochondria or chloroplasts. Instead, prokaryotes utilise specialised extensions of their plasma membrane (like mesosomes) for cellular respiration and energy production.
- **Statement D is incorrect:** A cell wall is not a universal feature of all cells. While plants, fungi, and most bacteria have a rigid cell wall, animal cells completely lack one, possessing only a flexible, semipermeable plasma membrane.

Q 81. Ans. C

Explanation:

- **Statement 1 is correct:** Proposed by Singer and Nicolson in 1972, the Fluid Mosaic Model is the universally accepted description of the cell membrane. The foundation of the membrane is a phospholipid bilayer (a double layer of fats). This lipid layer is not solid or rigid; rather, it behaves like a viscous fluid, allowing the membrane to be flexible and dynamic.
- **Statement 2 is correct:** The cell membrane is not made exclusively of fats. Scattered throughout this “sea” of fluid lipids are various proteins, resembling a mosaic pattern. These embedded proteins act like gatekeepers, channels, and pumps to help transport vital nutrients, water, and waste products in and out of the cell.

Q 82. Ans. B

Explanation:

- **Statement 1 is correct:** AlphaFold (developed by Google DeepMind) earned developers Demis Hassabis and John Jumper a Nobel prize for its ability to predict complex protein structures and how they fold from their amino acid sequences.
- **Statement 2 is incorrect:** While BioEmu-1 is developed by Microsoft, its purpose is to predict the full range of shapes a protein naturally explores under biological conditions. The model that predicts the impact of DNA mutations and categorises variants is AlphaGenome (and its complementary model AlphaMissense).
- **Statement 3 is incorrect:** AlphaGenome was introduced by Google DeepMind alone to predict how DNA mutations affect gene regulation.
 - ✓ The model designed to understand the language of individual cells and biological interactions is C2S-SCALE, developed by Google DeepMind and Yale University.

Q 83. Ans. B

Explanation:

- Mpox is a viral zoonotic disease caused by the monkeypox virus. Despite its misleading name, non-human primates are not the primary natural reservoir. Extensive ecological studies indicate that various species of small African rodents, such as rope squirrels, tree squirrels, and Gambian pouched rats, serve as the main reservoir hosts that sustain the virus in the wild.
- Monkeys were simply the first animals in which the disease was discovered during a laboratory outbreak in 1958, which is how the virus got its name. However, they are considered incidental hosts, much like humans, rather than the natural biological reservoir.
- Bats are the natural reservoirs for several highly lethal zoonotic viruses, including Nipah, Ebola, and Rabies. They are not associated with the transmission cycle of the monkeypox virus.
- Birds are primarily associated with the transmission of pathogens like avian influenza (Bird Flu) and the West Nile virus. They do not harbour or transmit mammalian orthopoxviruses.
- **Additional Facts:**
 - ✓ **Pathogen Family:** The monkeypox virus belongs to the Orthopoxvirus genus within the Poxviridae family. This is the exact same viral family that includes the variola virus, which historically caused Smallpox.
- In 2022, the World Health Organisation (WHO) officially recommended using the new synonym “Mpox”. This change was implemented to reduce the global stigma, discrimination, and racism associated with the original name.
- The virus is biologically divided into two distinct genetic clades: Clade I (formerly the Congo Basin clade, which is highly virulent and lethal) and Clade II (formerly the West African clade, which causes less severe illness).
- Because the viruses are closely related, epidemiological data shows that the traditional Smallpox vaccine provides approximately 85% cross-protective immunity against Mpox.

Q 84. Ans. B

Explanation:

- A GAN consists of two distinct neural networks, the Generator and the Discriminator, locked in a continuous game. The Generator tries to create fake data (like a fake photograph of a human face),

while the Discriminator acts as a detective, trying to guess if the image is real or fake. This competition forces the Generator to become incredibly good at creating hyper-realistic synthetic outputs.

● **Additional facts:**

- ✓ Deepfakes: GANs are the primary underlying technology used to create “Deepfakes”, highly realistic but entirely synthetic videos or audio recordings of real people saying or doing things they never did.
- ✓ Zero-Sum Game: The architecture is based on game theory. The Generator only “wins” when it successfully fools the Discriminator.
- ✓ Applications Beyond Media: In biotechnology, GANs are now being used to design entirely new molecular structures for drug discovery by predicting how synthetic proteins will fold and interact.

Q 85. Ans. C

Explanation:

- **Chloroplasts** are specialised plastids found primarily in the mesophyll cells of plant leaves. They contain the green pigment chlorophyll, which is essential for trapping solar energy (light energy). During the light-dependent reactions of photosynthesis, this trapped solar energy is utilised to synthesise energy-rich chemical molecules, specifically ATP (Adenosine Triphosphate) and NADPH. This specific process of creating ATP using light energy is known as photophosphorylation.
- **Mitochondria** are the “powerhouses” of the cell and generate the vast majority of cellular ATP, they do so through cellular respiration (oxidative phosphorylation) by breaking down glucose, not by capturing solar energy.
- **Lysosomes** are membrane-bound vesicular structures containing rich hydrolytic enzymes (hydrolases). They are primarily responsible for intracellular digestion and clearing cellular waste, often referred to as the “suicide bags” of the cell.
- **Endoplasmic Reticulum (ER)** is a complex network of tubular membranes. Rough ER (with ribosomes) is involved in protein synthesis and secretion, while Smooth ER is involved in lipid synthesis. It does not play a role in converting solar energy to ATP.

Q 86. Ans. C

Explanation:

- The Human Papillomavirus (HPV) is the most common sexually transmitted viral infection

globally. While many strains cause benign warts, persistent infection with “high-risk” strains (specifically HPV 16 and 18) is the primary biological cause of almost all cases of cervical cancer. The cervix is the lower, narrow end of the uterus that opens into the vagina.

● **Additional Facts:**

- ✓ CERVAVAC: This is India’s first indigenously developed Quadrivalent Human Papillomavirus (qHPV) vaccine. It was developed by the Serum Institute of India (SII) in Pune, making cervical cancer prevention significantly more affordable.
- ✓ Public Health Burden: Cervical cancer is the second most frequent cancer among women in India, despite being one of the most preventable forms of cancer through timely vaccination and regular Pap smear screenings.
- ✓ Target Demographic: The World Health Organisation (WHO) and the Government of India heavily recommend administering the HPV vaccine to pre-adolescent girls, ideally between the ages of 9 and 14 years, before any potential exposure to the virus.

Q 87. Ans. A

Explanation:

- **Option A is correct:** Peripheral immune tolerance is a crucial secondary safeguard mechanism of the human immune system. It takes place outside the primary lymphoid organs, specifically in the peripheral tissues and lymph nodes. Its primary function is to neutralise mature, self-reactive T-cells and B-cells that mistakenly escaped initial screening. It achieves this by forcing these rogue cells into a state of permanent unresponsiveness (anergy), triggering their self-destruction (apoptosis), or suppressing them using specialised Regulatory T-cells (Tregs).
- **Option B is incorrect:** This describes Central Immune Tolerance. Central tolerance is the primary screening process that occurs during the initial maturation of lymphocytes directly within the thymus (for T-cells) and the bone marrow (for B-cells).
- **Option C is incorrect:** This describes the fundamental action of the Innate Immune System, specifically phagocytosis. Macrophages and neutrophils act as the body’s first line of defence against external pathogens, which has nothing to do with distinguishing “self” from “non-self” tissues.

- **Option D is incorrect:** This describes the exact opposite of immune tolerance. The uncontrolled production of autoantibodies against healthy host tissue is the defining pathological mechanism of Autoimmune Diseases.

- **Additional Facts:**

- ✓ **Regulatory T-cells (Tregs):** These are a highly specialised subpopulation of T-cells. Their exclusive biological function is to suppress immune responses of other cells, acting as the “brakes” of the immune system to maintain peripheral tolerance.
- ✓ **Autoimmune Disorders:** When peripheral tolerance completely fails, the immune system begins attacking the body’s own organs. This directly results in severe autoimmune diseases such as Rheumatoid Arthritis, Type 1 Diabetes, and Systemic Lupus Erythematosus (SLE).
- ✓ **Cancer Evasion:** Tumours often hijack the mechanism of peripheral immune tolerance to survive. Cancer cells frequently recruit Tregs or express specific inhibitory proteins (like PD-L1) to artificially create a zone of tolerance, effectively blinding the immune system to the growing tumour.

Q 88. Ans. A

Explanation:

- **Statement 1 is correct:** Unlike most other organelles in a eukaryotic cell, mitochondria and chloroplasts possess their own independent genetic machinery. They contain double-stranded, circular DNA molecules, RNA, and their own ribosomes (70S type, which is similar to bacterial ribosomes). This unique feature strongly supports the “Endosymbiotic Theory,” which suggests that these organelles were once free-living bacteria that were engulfed by a larger host cell millions of years ago.
- **Statement 2 is incorrect:** While these organelles do have their own DNA and can synthesise some of their required proteins, they cannot synthesise all of them. They remain heavily dependent on the DNA located within the cell’s main nucleus for the majority of the proteins and enzymes required for their structure and function. Because of this partial dependence, they are accurately classified as semi-autonomous organelles, rather than fully autonomous ones.

Q 89. Ans. A

Explanation:

- **Assertion (A) is true:** Artificial trans fats are universally recognised by the World Health

Organisation (WHO) as the most dangerous type of dietary fat. Due to their direct link to fatal cardiovascular diseases, global health agencies, including India’s FSSAI, have implemented strict regulations to phase them out of the commercial food supply completely.

- **Reason (R) is true and explains (A):** The extreme harmfulness of trans fats stems directly from their unique chemical structure. They are primarily created through a prolonged industrial process called “partial hydrogenation,” where hydrogen gas is forced into liquid vegetable oils to make them solid at room temperature (like margarine or vanaspati ghee). Biologically, these trans fats are dangerous because they perform a dual negative action: they significantly increase Low-Density Lipoprotein (LDL or “bad” cholesterol) while simultaneously decreasing High-Density Lipoprotein (HDL or “good” cholesterol). Therefore, the Reason perfectly explains the public health Assertion.

- **Additional facts:**

- **Trans Fats (Artificial & Natural):**
 - ◇ **Sources:** Commercially baked goods, deep-fried street foods, vanaspati ghee, and margarine. (Trace amounts naturally occur in ruminant meat and dairy).
 - ◇ They strictly serve to increase the shelf-life of commercial food products.
 - ◇ **Problems:** The most dangerous fat. Causes severe systemic inflammation, insulin resistance (Type 2 Diabetes), and is a primary driver of coronary artery disease and heart attacks.
- **Saturated Fats:**
 - ◇ **Sources:** Animal-based products (butter, desi ghee, cheese, red meat) and tropical plant oils (coconut oil, palm oil). They are typically solid at room temperature.
 - ◇ **Benefits:** Required in small amounts for cellular structure and hormone production.
 - ◇ **Problems:** Overconsumption can elevate total blood cholesterol levels and LDL (bad) cholesterol, potentially increasing the risk of blockages in arteries. Public health guidelines recommend moderate consumption.
- **Monounsaturated Fats (MUFAs):**
 - ◇ **Sources:** Olive oil, mustard oil,

avocados, almonds, peanuts, and cashews. They are liquid at room temperature but turn cloudy/solid when chilled.

- ◇ Benefits: Highly beneficial. They actively help lower LDL cholesterol without affecting HDL cholesterol, reduce cellular inflammation, and maintain a healthy cardiovascular system.
- Polyunsaturated Fats (PUFAs):
 - ◇ Sources: Sunflower oil, safflower oil, walnuts, flaxseeds, and fatty fish (like salmon and mackerel).
 - ◇ Benefits: Highly beneficial and essential. This group includes Omega-3 and Omega-6 fatty acids, which the human body cannot synthesise on its own. They are critical for brain function, building cell membranes, blood clotting, and fighting systemic inflammation.

Q 90. Ans. A

Explanation:

- **Option A is correct:** Semaglutide belongs to a class of medications known as GLP-1 (Glucagon-like peptide-1) receptor agonists. The human body naturally produces the GLP-1 hormone in the intestines after eating. This medication mimics that exact hormone. It travels to the brain and activates receptors that regulate appetite, making the person feel full much faster and for a longer duration. Additionally, it slows down gastric emptying, meaning food stays in the stomach longer, which drastically reduces overall caloric intake.
- **Option B is incorrect:** This describes the mechanism of older weight-loss drugs like Orlistat. Orlistat works locally in the gut as a lipase inhibitor, preventing the intestines from absorbing about a third of the fat in the food consumed.
- **Option C is incorrect:** This describes the mechanism of stimulant-based appetite suppressants like Phentermine. These drugs trigger the release of neurotransmitters like norepinephrine in the brain, inducing a “fight or flight” response that artificially boosts metabolism and suppresses hunger, often with severe cardiovascular side effects.
- **Option D is incorrect:** This describes the function of diuretics (water pills). While they cause rapid weight loss on the scale by forcing the kidneys to

excrete sodium and water, they do not reduce actual body fat and are extremely dangerous if misused for weight management.

● Additional Facts:

- ✓ The Danish pharmaceutical giant Novo Nordisk has recently launched this breakthrough weight-loss drug in India under the brand name Wegovy.
- ✓ Drugs like Semaglutide (also sold under the brand name Ozempic) were originally developed and approved strictly for managing Type 2 Diabetes because they stimulate the pancreas to release insulin when blood sugar levels are high.
- ✓ The introduction of these drugs is highly relevant to India’s public health policy, as the country is currently facing a massive epidemic of Non-Communicable Diseases (NCDs), with obesity acting as the primary catalyst for diabetes and cardiovascular diseases.

Q 91. Ans. B

Explanation:

- In 1953, James Watson and Francis Crick proposed the double-helix model for the structure of DNA. However, they did not conduct the foundational laboratory experiments themselves. Their model was primarily based on the vital X-ray crystallography (diffraction) data produced by **Rosalind Franklin** and **Maurice Wilkins**. Franklin’s famous “Photo 51” explicitly revealed the helical nature and fundamental dimensions of the DNA molecule. For this discovery, Watson, Crick, and Wilkins were awarded the Nobel Prize in Medicine in 1962 (Rosalind Franklin had passed away, and Nobel prizes are not awarded posthumously).
- The discovery of DNA polymerase, the enzyme responsible for synthesising DNA, was made by **Arthur Kornberg** in 1956, after the double-helix structure had already been proposed.
- The **Hershey-Chase experiment (1952)** used radioactive phosphorus and sulfur to definitively prove that DNA, not protein, was the genetic material entering cells. While historically crucial, it did not reveal the physical structure or shape of the DNA molecule.
- The **Human Genome Project** was an international scientific research project that was launched in 1990 and completed in 2003. It mapped the sequence of nucleotide base pairs.

Q 92. Ans. B**Explanation:**

- During the process of translation (protein synthesis), the cellular machinery “reads” the mRNA sequence in sets of three nucleotides called **codons**. Out of the 64 possible codons, 61 specifically code for different amino acids.
- However, three specific codons, **UAA, UAG, and UGA**, do not code for any amino acids. Because they do not specify an amino acid, they are referred to as “nonsense” codons. Their primary biological function is to act as a “**stop**” signal. When the ribosome encounters any of these three codons, it terminates the translation process and releases the newly formed protein chain into the cell.
- **AUG** is actually the universal “**start**” or initiation codon, which also codes for the amino acid **Methionine**.

Q 93. Ans. B**Explanation:**

- mRNA stands for Messenger Ribonucleic Acid.
- Its primary function is to act as a crucial intermediary in the process of making proteins.
- The DNA, which contains the master genetic blueprint, is safely housed inside the cell’s nucleus. However, the actual “factories” that build proteins, the ribosomes, are located outside the nucleus in the cytoplasm.
- mRNA is created by copying a specific segment of DNA (a process called transcription). It then carries this genetic “message” out of the nucleus to the ribosomes, dictating the exact sequence of amino acids needed to build a specific protein (a process called translation).

Q 94. Ans. C**Explanation:**

- **Pair 1 is correctly matched:** Type 1 Diabetes (formerly known as juvenile-onset diabetes) is strictly an autoimmune disease. The body’s own immune system destroys the beta cells in the pancreas. Consequently, the body produces absolutely zero insulin, making lifelong external insulin administration mandatory for survival.
- **Pair 2 is correctly matched:** Type 2 Diabetes (formerly adult-onset diabetes) is the most common form globally. It is heavily driven by lifestyle factors, genetics, and obesity. The body still produces insulin, but the liver, muscle, and fat cells become

“resistant” to it. Over time, the pancreas cannot make enough insulin to overcome this resistance.

- **Pair 3 is correctly matched:** Gestational Diabetes occurs when pregnancy hormones interfere with the body’s ability to use insulin effectively, leading to temporary insulin resistance. While blood sugar levels usually return to normal shortly after delivery, it significantly increases the mother’s future risk of developing Type 2 Diabetes.
- **Pair 4 is incorrectly matched:** Diabetes Insipidus has absolutely nothing to do with blood sugar (glucose) or dietary sugar intake. It is a completely separate and rare disorder of fluid regulation. It is caused by a deficiency of, or insensitivity to, the antidiuretic hormone (ADH or vasopressin), which leads to extreme thirst and massive volumes of dilute urine.
- **Additional Facts:**

- ✓ **HbA1c Test:** This is the gold-standard diagnostic blood test for diabetes. It measures the percentage of blood sugar attached to haemoglobin, providing an average of the patient’s blood glucose levels over the past 2 to 3 months.
- ✓ **The Role of Insulin:** Insulin is a peptide hormone produced by the pancreas. Its primary biological function is to unlock cells, allowing glucose from the bloodstream to enter and be used for cellular energy.
- ✓ **Public Health Crisis:** India is frequently referred to as the “Diabetes Capital of the World.” The prevalence of Type 2 Diabetes is escalating rapidly, driven by rapid urbanization, shifting dietary patterns towards processed foods, and increasingly sedentary lifestyles.

Q 95. Ans. C**Explanation:**

- **Statement 1 is correct:** Human somatic (body) cells are diploid, meaning they contain 46 chromosomes (23 pairs). If gametes (sperm and egg) were produced through regular cell division (mitosis), they would also contain 46 chromosomes. Upon fertilisation, the resulting offspring would end up with 92 chromosomes, fundamentally altering the species. Meiosis solves this by acting as a reduction division. It reduces the chromosome number by exactly half, creating haploid gametes containing only 23 single chromosomes. When a sperm and egg fuse during fertilisation, the normal diploid number of 46 is perfectly restored in the zygote.

- **Statement 2 is correct:** Meiosis is the primary biological engine of genetic diversity in sexually reproducing organisms. During the first phase of meiosis (specifically Prophase I), paired homologous chromosomes align closely together and physically swap segments of their DNA. This phenomenon is known as “crossing over” or genetic recombination. Because of this exchange, the four resulting gametes are genetically unique from one another and from the parent cell. This ensures that no two children from the same parents are exactly alike (excluding identical twins), providing a critical foundation for evolutionary adaptation and survival.

Q 96. Ans. B

Explanation:

- **Statement 1 is incorrect:** While it is a fully indigenous and highly automated Air Defence Control and Reporting System, it was strictly developed by Bharat Electronics Limited (BEL), but DRDO and ISRO also contributed to its development.
- **Statement 2 is correct:** Akashteer is an integral part of the modern C4ISR (Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance) framework. It is specifically designed as a vehicle-based system. This high mobility allows the Indian Army to rapidly deploy and reposition the control centres during active combat.
- **Statement 3 is incorrect:** Project Akashteer reflects the exact opposite strategic shift. By leveraging AI to seamlessly link various radar nodes and firing units, it enables a shift from passive, isolated defence monitoring to highly coordinated, proactive retaliation against hostile drones, rockets, and aircraft.

Q 97. Ans. C

Explanation:

- **Option A is incorrect:** This statement describes Mendel’s Law of Segregation (also known as the purity of gametes). It explains that allele pairs separate or segregate during gamete formation, meaning a sperm or egg carries only one allele for each inherited trait.
- **Option B is correct:** Mendel’s Law of Independent Assortment states that when two or more pairs of traits are considered, the segregation of one pair of traits is entirely independent of the other pair of traits during the formation of gametes.

For example, the gene determining a pea plant’s seed colour (yellow or green) is passed on to the offspring independently of the gene determining its seed shape (round or wrinkled). Mendel derived this principle from his famous dihybrid cross experiments.

- **Option C is incorrect:** This statement describes the Law of Dominance, which explains why some traits are expressed over others in a hybrid organism.
- **Option D is incorrect:** This statement describes Genetic Linkage. Linkage was discovered later by Thomas Hunt Morgan and is an exception to Mendel’s Law of Independent Assortment. Genes located very close together on the same chromosome tend to be inherited together and do not assort independently.

Q 98. Ans. C

Explanation:

- **Option A is a component:** The Quick Reaction Surface-to-Air Missile (QRSAM) is an indigenous short-range SAM developed directly by the DRDO. It is uniquely configured on a highly mobile 6x6 truck platform. Its specific tactical purpose is to provide an active, moving umbrella of air defence over advancing Indian Army armoured columns.
- **Option B is a component:** The Advanced Very Short-Range Air Defence System (VSHORADS) is developed by the Research Centre Imarat (RCI), a premier DRDO laboratory. It is a Man-Portable Air Defence System (MANPADS). This means it can be carried and fired by individual soldiers from their shoulders to quickly neutralise low-altitude, close-range aerial threats like drones and attack helicopters.
- **Option C is not a component:** The Helina (Helicopter-launched Nag) is an Anti-Tank Guided Missile (ATGM). It is strictly an air-to-surface weapon designed to destroy enemy tanks and armoured vehicles. It has absolutely no role in an air defence (surface-to-air) framework.
- **Option D is a component:** A high-power laser-based Directed Energy Weapon (DEW) developed by the Centre for High Energy Systems and Sciences (CHESS). DEWs represent the next generation of warfare, utilising concentrated laser beams rather than kinetic projectiles to silently burn through or blind incoming drones and missiles.
- **Additional Facts:**
 - ✓ Dual Propulsion in VSHORADS: The Indian VSHORADS missile incorporates a

state-of-the-art dual-thrust solid motor, giving it the necessary speed and manoeuvrability to hit fast-moving targets.

- ✓ The QRSAM system is stored and launched directly from environmentally sealed canisters. This drastically reduces maintenance requirements and allows the missile to be kept in a state of continuous readiness for immediate firing without complex setup.
- ✓ The Centre for High Energy Systems and Sciences (CHESS) is situated in Hyderabad. It is the nodal DRDO laboratory exclusively tasked with developing cutting-edge directed energy, laser, and microwave weapon systems for the Indian armed forces.

Q 99. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Destroyers are the heavily armed workhorses of modern blue-water navies. They are fast, highly manoeuvrable escorts that provide a comprehensive defensive “shield” (Anti-Air, Anti-Submarine, and Anti-Surface) around highly valuable, vulnerable assets like Aircraft Carriers.
- **Pair 2 is incorrectly matched:** The description provided belongs to Amphibious Assault Ships (such as Landing Helicopter Docks - LHDs), not Cruisers. Cruisers are the largest and most powerful traditional surface combatant ships in active service today (excluding aircraft carriers). They are capable of operating completely independently or leading their own maritime task forces.
- **Pair 3 is correctly matched:** Frigates are a step below destroyers in size, armament, and speed. While modern stealth frigates possess multi-role capabilities, their foundational naval doctrine has always prioritised Anti-Submarine Warfare (ASW) and escorting unarmed merchant fleets through hostile waters.
- **Pair 4 is correctly matched:** Corvettes are the smallest recognised class of true warships. Because they carry less fuel and fewer supplies, they lack the extreme endurance of frigates or destroyers. Therefore, they are optimised for “brown-water” or littoral (coastal) operations, fast strikes, and policing a nation’s Exclusive Economic Zone (EEZ).
- **Additional Fact:**
 - ✓ To understand modern naval fleets, surface combatants generally follow this tonnage/size hierarchy (from largest to smallest): Cruiser >

Destroyer > Frigate > Corvette > Fast Attack Craft/Patrol Boat.

- ✓ Project 15B (Visakhapatnam Class): This is the Indian Navy’s latest and most lethal class of indigenous stealth guided-missile Destroyers, built by Mazagon Dock Shipbuilders Limited (MDL). It includes INS Visakhapatnam, INS Mormugao, INS Imphal, and INS Surat.
- ✓ Project 17A (Nilgiri Class): This represents India’s ongoing program to construct highly advanced, radar-evading stealth Frigates.
- ✓ Project 28 (Kamorta Class): These are India’s indigenously built Anti-Submarine Warfare (ASW) Corvettes. Because their primary job is hunting submarines, they are specifically built using special high-tensile steel to minimise their underwater acoustic signature (noise).

Q 100. Ans. D

Explanation:

- Nanotechnology is highly interdisciplinary, and all five of the listed fields heavily utilise nanoscale materials (typically between 1 and 100 nanometers) to achieve revolutionary results.
- Targeted drug delivery (Medicine): Nanoparticles are engineered to carry pharmaceutical drugs directly to diseased cells (such as targeted chemotherapy for cancer tumours), which drastically minimises toxic side effects on surrounding healthy tissues.
- Heavy metal water purification (Environment): Nanomaterials, particularly carbon nanotubes and magnetic nanoparticles, are highly effective at absorbing, filtering, and neutralising toxic heavy metals and microscopic pathogens from industrial wastewater.
- Quantum dot displays (Electronics): Modern QLED televisions and monitors utilise nanocrystals known as “quantum dots” that emit specific, highly vibrant colours when illuminated, resulting in superior image quality and energy efficiency.
- Self-cleaning fabrics (Textiles): Everyday textiles and industrial fabrics coated with nanoparticles (like titanium dioxide or zinc oxide) become highly hydrophobic. They actively repel water, resist deep stains, and kill odour-causing bacteria.
- Smart packaging for agriculture (Food/Agri): Nanosensors integrated directly into food packaging can actively monitor the environment, detect early bacterial growth, and alert consumers to spoilage, vastly extending shelf life and reducing food waste.
- **Additional Facts:**

- ✓ Graphene: This is one of the most famous nanomaterials. It is a single layer of carbon atoms arranged in a 2D hexagonal lattice. It is highly prized for being roughly 200 times stronger than steel while remaining highly flexible and an exceptional conductor of electricity.
- ✓ India's Nano Mission: To promote domestic research and capacity building, the Government of India launched the umbrella "Nano Mission" (National Mission on Nano Science and Technology) in 2007, spearheaded by the Department of Science and Technology (DST).
- ✓ The "Bottom-Up" Approach: Nanomanufacturing often uses a bottom-up approach, where complex materials and devices are built molecule by molecule, or atom by atom, allowing for unprecedented structural precision.

Q 101. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Haemophilia is a classic X-linked recessive disorder. Because it is carried on the X chromosome, it predominantly affects males (who have only one X chromosome), while females act as carriers unless both of their X chromosomes carry the defective gene.
- **Pair 2 is correctly matched:** Sickle-cell anaemia is an autosomal recessive trait. It is controlled by a single pair of alleles on the autosomes (non-sex chromosomes). It can be transmitted from parents to offspring when both partners are carrier for the gene (heterozygous).
- **Pair 3 is incorrectly matched:** Down syndrome is not a monosomy. It is caused by the presence of an additional copy of chromosome number 21. This condition is scientifically referred to as Trisomy 21.
- **Pair 4 is incorrectly matched:** Klinefelter's syndrome is caused by the presence of an additional copy of the X chromosome in males, resulting in a karyotype of 47 chromosomes (44 Autosomes + XXY). The genetic makeup of "44 Autosomes + XO" (absence of one X chromosome) actually describes Turner's syndrome in females.

Q 102. Ans. A

Explanation:

- **Statement 1 is correct:** Second-generation fighters (developed in the 1950s and 60s) broke the sound barrier in level flight. This generation introduced

airborne radar systems and the first infrared-guided missiles (like the AIM-9 Sidewinder). Example: MiG-21.

- **Statement 2 is correct:** Third-generation fighters (1960s and 70s) were designed as multi-role aircraft capable of both air superiority and ground attack. They heavily utilised improved radar to fire BVR (Beyond Visual Range) missiles. Example: F-4 Phantom and MiG-23.
- **Statement 3 is correct:** Fourth-generation fighters (1970s to 1990s) combat with "fly-by-wire" computers. Instead of mechanical pulleys, computers interpreted the pilot's inputs, allowing jets to be aerodynamically unstable but incredibly agile and manoeuvrable. Examples: F-16, Su-27, Mirage 2000.
- **Statement 4 is incorrect:** Fifth-generation fighters do not rely on ramjets for hypersonic flight. The defining characteristics of a 5th-generation jet are Stealth (Very Low Observable radar cross-section), Sensor Fusion (computers combining all sensor data into a single battlespace picture), and Supercruise (the ability to fly at supersonic speeds without using fuel-guzzling afterburners). Examples: F-22 Raptor, F-35 Lightning II, Su-57, and China's J-20.

Additional Facts:

- ✓ The "4.5 Generation": Because 5th-generation stealth jets are overwhelmingly expensive to develop and maintain, many air forces rely on 4.5-generation jets. These are essentially highly upgraded 4th-gen airframes fitted with 5th-gen electronics, specifically AESA (Active Electronically Scanned Array) radars.
- ✓ The Indian Air Force's Rafale and the indigenous LCA Tejas Mk-1A perfectly represent 4.5-generation capabilities.
- ✓ Project AMCA: India is actively developing its own indigenous 5th-generation stealth fighter under the Advanced Medium Combat Aircraft (AMCA) program, spearheaded by the DRDO and HAL.
- ✓ First Generation: For context, 1st-generation fighters were the earliest subsonic jet aircraft of the Korean War era, entirely lacking radar and relying solely on unguided machine guns and cannons. Example: F-86 Sabre and MiG-15.

Q 103. Ans. B

Explanation:

- The Hyperloop is a proposed ultra-high-speed ground transportation system. It relies on two

primary physical principles. First, it uses magnetic levitation (maglev) to float the passenger pods, completely eliminating rolling friction with the track. Second, the pods travel through a sealed network of tubes where powerful pumps remove almost all the air to create a near-vacuum (low-pressure environment). This virtually eliminates aerodynamic drag (air resistance), allowing the pods to glide at airline speeds (up to 1,000+ km/h) using very little electrical energy.

- This describes the fundamental operational mechanics of a standard commercial jet airliner flying in the upper atmosphere, not a ground-based tube transport system.
- This describes a traditional pneumatic tube system (like the ones historically used in banks or post offices to send small capsules). Hyperloop uses a vacuum (pulling air out) rather than positive pressure (pushing air in) to achieve its speeds.
- While superconducting cables are used in advanced power grids and some maglev systems, simply carrying high-voltage current does not describe the physical propulsion and drag-reduction mechanism that defines the Hyperloop.
- **Additional Facts:**
 - ✓ The “Fifth Mode” of Transport: The modern Hyperloop concept was first proposed in a 2013 white paper by Elon Musk (CEO of SpaceX and Tesla), who famously described it as the “fifth mode of transport” alongside planes, trains, cars, and boats.
 - ✓ NITI Aayog has actively explored the technological and commercial viability of Hyperloop in India. Previously, Virgin Hyperloop had proposed a flagship route connecting Mumbai and Pune, which aimed to reduce the 3-hour journey to roughly 25 minutes.
 - ✓ Because the system overcomes the two biggest barriers to high-speed travel (air resistance and ground friction), it requires a fraction of the energy per passenger compared to a jet aircraft. If the system is powered by solar panels mounted along the top of the tubes, it could potentially be a completely carbon-neutral transit system.

Q 104. Ans. C

Explanation:

- **Ribosomes** are dense, granular structures found in all living cells, including both prokaryotes

and eukaryotes. They serve as the indispensable “protein factories” of the cell. During the process of translation, ribosomes attach to messenger RNA (mRNA) and read its genetic instructions. They then facilitate the linking of specific amino acids in the correct sequence to form long polypeptide chains, which ultimately fold into functional proteins.

- **Golgi apparatus** functions as the cellular post office. It is primarily responsible for modifying, sorting, and packaging proteins and lipids for delivery to other targeted destinations.
- **Mitochondria** are the powerhouses of the cell, responsible for generating cellular energy in the form of ATP through the process of cellular respiration.
- **Smooth Endoplasmic Reticulum (SER)** completely lacks ribosomes on its surface. It is primarily involved in the synthesis of lipids (fats), steroid hormones, and the detoxification of harmful chemicals, rather than protein synthesis.

Q 105. Ans. D

Explanation:

- The Black Box (which actually consists of the Flight Data Recorder and Cockpit Voice Recorder) is heavily armoured to survive crashes and records all mechanical flight parameters and cockpit audio for accident investigation. It is painted bright orange and equipped with an underwater locator beacon.
- A Ground Proximity Warning System (GPWS) is an active instrument that alerts pilots if the aircraft is in immediate danger of flying into the ground.
- Traffic Collision Avoidance System (TCAS) communicates with the transponders of other nearby aircraft to prevent mid-air collisions.
- The Flight Management Computer (FMC) is the “brain” of the navigation system that automates in-flight tasks, but it is not armoured to survive a crash or record flight history for investigations.

Q 106. Ans. B

Explanation:

- **Altermagnetism** is a recently confirmed third fundamental class of magnetism. It essentially combines the best features of the two traditional classes. Like antiferromagnets, altermagnets have no net macroscopic magnetic field (meaning they don’t stick to the fridge and don’t create stray fields). However, due to their unique crystal symmetries, they still exhibit strong “spin-polarised” electron bands, a useful property previously thought to belong almost exclusively to ferromagnets

- **Additional Facts:**

- ✓ The Spintronics Revolution: Standard electronics use the charge of an electron to process data. "Spintronics" aims to use the spin of an electron to create vastly faster and more energy-efficient computers. Altermagnets are considered a "holy grail" for spintronics because their lack of a net magnetic field means tightly packed computer chips won't interfere with each other, while their spin-polarisation allows data to be carried effectively.
- ✓ While theorised a few years earlier, the existence of altermagnets was definitively proven in early 2024 by international researchers using advanced X-ray spectroscopy at facilities like the Swiss Light Source (SLS).
- ✓ Candidate Materials: Common materials proven to exhibit altermagnetism include Manganese telluride and Ruthenium dioxide.

Q 107. Ans. C

Explanation:

- **Glyphosate** is a highly controversial, broad-spectrum synthetic herbicide. Its primary biological function is to kill unwanted weeds and grasses by inhibiting a specific plant enzyme pathway. It does not stimulate plant growth or improve crop health.
- **Humic and fulvic acids** are complex organic molecules found naturally in soil. When applied to crops, they act as powerful biostimulants by significantly improving the soil structure, increasing root mass, and enhancing the plant's ability to absorb vital nutrients.
- **Seaweed extracts** (commonly derived from brown algae like *Ascophyllum nodosum*) are heavily used as biostimulants. They are rich in natural plant hormones (like auxins and cytokinins) that actively promote cellular division and build the plant's resilience against abiotic stresses like extreme heat or drought.
- **Arbuscular Mycorrhizal Fungi (AMF)** are highly beneficial soil microorganisms. They form a symbiotic relationship with plant roots, effectively acting as an extension of the root system to help the plant scavenge for water and difficult-to-absorb nutrients like phosphorus.
- **Additional Facts:**
 - ✓ In a major agricultural reform, the Government of India amended the Fertiliser (Inorganic, Organic or Mixed) (Control) Order (FCO) in 2021 to officially recognise "Biostimulants" as

a distinct, regulated category of agricultural inputs for the first time.

- ✓ It is crucial for regulatory and examination purposes to understand that biostimulants are entirely distinct from Fertilizers (which directly provide bulk nutrients like NPK) and Pesticides (which directly kill pests or diseases). Biostimulants operate by enhancing the plant's natural physiological processes.
- ✓ The promotion of biostimulants is a core component of India's push towards natural farming and sustainable agriculture, aiming to reduce the excessive reliance on harmful chemical fertilisers and boost long-term soil health.

Q 108. Ans. D

Explanation:

- Lysosomes act as the primary waste disposal system of the cell. They are membrane-bound sacs filled with powerful digestive enzymes (hydrolytic enzymes) capable of breaking down complex organic materials. They help keep the cell clean by actively digesting any foreign material that enters the cell, such as harmful bacteria or food particles, as well as breaking down old, poorly functioning cellular organelles.
- During a disturbance in cellular metabolism, for example, when the cell gets damaged, lysosomes may burst, and the enzymes digest their own cell. Therefore, lysosomes are also popularly known as the "suicide bags" of a cell.

Q 109. Ans. B

Explanation:

- **Osteology** (derived from the Greek words osteon, meaning "bone", and logos meaning "study") is the specialised branch of gross anatomy, physical anthropology, and palaeontology dedicated to the scientific study of bones, skeletal elements, teeth, microbone morphology, and the ossification process.
- **Myology:** This is the scientific study of the muscular system, including the structure, function, and diseases of muscles.
- **Cytology:** This is the branch of biology concerned with the study of the structure, chemistry, and function of individual plant and animal cells.
- **Histology:** Also known as microscopic anatomy, this is the study of the microscopic structure of biological tissues.

● Additional Facts:

- ✓ Bones are not static; they are highly dynamic living tissues continuously being remodelled. This process is governed primarily by two types of cells: Osteoblasts (which actively build new bone tissue) and Osteoclasts (which break down and resorb old or damaged bone).
- ✓ Bone Composition: Chemically, bone tissue is a composite material made of an organic matrix (primarily flexible collagen fibres) and inorganic mineral salts (primarily calcium phosphate in a crystalline arrangement called hydroxyapatite), which provides immense compressive strength.

Q 110. Ans. C

Explanation:

- In an Earthworm (an annelid), the respiratory pigment hemoglobin is not contained within specialized red blood cells (RBCs) as it is in vertebrates. Instead, the hemoglobin is dissolved directly into the blood plasma.

Q 111. Ans. B

Explanation:

- Anopheles mosquitoes are morphologically distinguished in the field by their unique resting posture. When they land on a surface, adults rest with their heads down and their abdomens pointing upward at a distinct angle (often around 45 degrees). In contrast, most other common mosquito genera, including Culex and Aedes, rest with their bodies positioned roughly parallel to the surface.

Q 112. Ans. C

Explanation:

- Unlike plants, fungi entirely lack **chlorophyll**. Because they do not possess this pigment, they cannot perform photosynthesis to produce their own food (autotrophy). Instead, fungi are heterotrophs, meaning they must absorb their nutrients from other living or dead organic matter.

Q 113. Ans. D

Explanation:

- **Statement 1 is correct:** The Endoplasmic Reticulum is a large network of membrane-bound tubes and sheets. The Rough ER (RER) looks rough under an electron microscope because it has particles called ribosomes attached to its surface. Since ribosomes are the “protein factories” of the cell, the RER

is fundamentally involved in synthesising and modifying proteins, which are then sent to various places in the cell depending on need.

- **Statement 2 is correct:** The Smooth ER (SER) does not have ribosomes attached to it, hence its smooth appearance. Its primary cellular function is the synthesis of essential lipids and steroid hormones. Furthermore, in the liver cells of vertebrates, the SER plays a vital, specialised role in detoxifying many harmful metabolic by-products, drugs, and external poisons.

Q 114. Ans. C

Explanation:

- An ecological niche refers to the specific functional role or “profession” an organism plays within its ecosystem. It encompasses everything an organism does to survive and reproduce, including what it eats, what eats it, its behaviour, how it alters the environment, and its interactions with both living (biotic) and non-living (abiotic) factors.
- A biome is a massive geographical area characterised by its climate and the dominant vegetation and animal life found there (e.g., Tundra, Tropical Rainforest, Desert).
- An ecotone is a transition zone or boundary area where two different biological communities or ecosystems meet and integrate (e.g., a marshland between a river and a forest, or an estuary where freshwater meets the ocean). Ecotones usually have high biodiversity.
- Ecological succession is the gradual, predictable process by which the structure of a biological community changes and evolves over time, often following a disturbance like a forest fire or volcanic eruption.
- Habitat vs. Niche: A classic ecological analogy is that a habitat is an organism’s “address” (where it lives), while a niche is its “profession” (what it does for a living).

Q 115. Ans. B

Explanation:

- A Lichen is not a single organism, but rather a remarkable mutualistic association between two distinct entities: a fungus (the mycobiont) and a photosynthetic partner, either an alga or a cyanobacterium (the phycobiont). In this symbiotic relationship, the fungus provides a protective physical structure, gathers moisture, and extracts minerals from the surrounding environment. In

return, the photosynthetic partner, whether an alga or a cyanobacterium, both of which contain chlorophyll and are capable of photosynthesis, produces carbohydrates that sustain both organisms.

Q 116. Ans. B

Explanation:

- The Golgi apparatus (or Golgi complex) functions very much like the “post office” or packaging and dispatch centre of the cell. Materials that are synthesised near the Endoplasmic Reticulum (such as proteins and lipids) are sent to the Golgi apparatus.
- Here, these materials are chemically modified, carefully sorted, and packaged into small membrane-bound sacs (vesicles). They are then dispatched to various target destinations either inside the cell or secreted outside the cell membrane.

Q 117. Ans. B

Explanation:

- **Mycorrhiza** (plural: mycorrhizae) refers to the mutually beneficial (symbiotic) association between a specialised soil fungus and the roots of a vascular plant. In this relationship, the fungal hyphae vastly extend the root system’s surface area, helping the plant absorb difficult-to-reach water and essential minerals (especially phosphorus). In return, the plant provides the fungus with carbohydrates (sugars) produced through photosynthesis.
- **Parasitism:** This is a relationship where one organism (the parasite) benefits at the direct expense and harm of the other organism (the host), such as a tick on a dog or a tapeworm in an intestine.
- **Saprophytism:** This refers to organisms (like many decomposer fungi and bacteria) that obtain their nutrients by breaking down dead and decaying organic matter, rather than interacting with living plant roots.
- **Commensalism:** This is an ecological interaction where one species benefits while the other is completely unaffected (neither helped nor harmed). An example is a bird building a nest in a large tree.

Q 118. Ans. C

Explanation:

- Decomposers (primarily bacteria and fungi) are organisms that break down complex, dead, and decaying organic matter into simpler inorganic substances like carbon, nitrogen, and phosphorus.

They secrete specialized enzymes into their environment to break down the organic material externally before absorbing the nutrients. This process cleans the environment and recycles essential nutrients back into the soil and atmosphere for plants to use again.

Q 119. Ans. D

Explanation:

- **Organoid intelligence (OI)** is a revolutionary multidisciplinary field that uses 3D cultures of human brain cells, known as brain organoids, as biological hardware for computing. Instead of relying on traditional rigid silicon microchips, OI seeks to harness the complex processing power, structural plasticity, and extreme energy efficiency of living biological neural networks to perform computational memory and learning tasks.

Q 120. Ans. A

Explanation:

- **Charles Darwin** is fundamentally associated with the concept of **Natural Selection**. He introduced this mechanism of evolution in his groundbreaking 1859 book, *On the Origin of Species*. Natural selection is the process by which organisms better adapted to their environment tend to survive and produce more offspring.
- **Inheritance of Acquired Characters:** Proposed by **Jean-Baptiste Lamarck**. This theory (often called Lamarckism) suggested that organisms could pass down physical traits acquired during their lifetime to their offspring (e.g., a giraffe stretching its neck to reach high leaves would pass a longer neck to its babies).
- **Mutation Theory:** Proposed by **Hugo de Vries**. He suggested that new species arise from sudden, large, and unpredictable genetic changes (mutations) rather than gradual, continuous variations.
- **Germplasm Theory:** Proposed by **August Weismann**. This theory states that heritable information is transmitted only by germ cells (sperm and egg) in the gonads, and that changes to somatic (body) cells cannot be passed on to offspring, effectively disproving Lamarckism.

Q 121. Ans. B

Explanation:

- **Muon tomography** (or muography) is a revolutionary imaging technique that uses naturally occurring muons, which are heavy, high-energy

subatomic particles created when cosmic rays collide with the Earth's upper atmosphere. Because muons are much heavier than electrons, they can easily pass through hundreds of meters of solid rock or dense metal. By placing detectors around a large structure to track how many muons pass through, scientists can create a 3D "X-ray" of its interior.

- **Carbon dating:** This is a radiometric dating method used to determine the exact age of organic materials (like ancient wood or bones) based on the decay of the Carbon-14 isotope. It does not image internal structures.
- **LiDAR scanning:** Light Detection and Ranging uses pulsed lasers to map external surface topography with high precision. It is excellent for finding ruins hidden under dense jungle canopies, but it cannot see through solid stone.
- **Seismic refraction:** This technique uses controlled acoustic shockwaves (like small explosives or heavy thumping) to map underground geological layers based on how sound waves bend.
- **Additional Facts:**
 - ✓ ScanPyramids Project: In 2017, muon tomography made global headlines when it was used to discover a massive, previously unknown hidden void (chamber) deep inside the Great Pyramid of Giza without moving a single stone.
 - ✓ Because the technique is entirely passive and doesn't require humans to enter dangerous areas, it was crucially deployed at the highly radioactive Fukushima Daiichi nuclear power plant in Japan to safely locate the melted nuclear fuel cores.
 - ✓ National Security: Modern border security agencies are increasingly exploring muon scanners for cargo ports to detect shielded nuclear contraband or heavy metals hidden inside shipping containers that standard X-rays cannot penetrate.

Q 122. Ans. B

Explanation:

- **Biomining** is a highly sustainable metallurgical technique that utilises specialised extremophile microorganisms (most commonly *Acidithiobacillus ferrooxidans*) to extract valuable metals like copper, gold, and uranium from low-grade ores. In a process called "bioleaching," these bacteria essentially feed on the sulfur and iron binding the rock, oxidising the minerals and dissolving the trapped metals into a liquid solution that can be easily collected and purified.

- **Bioprinting:** This involves using 3D printing technology equipped with "bio-ink" (living cells) to construct complex biological tissues or artificial organs layer by layer.
- **Biostimulation:** This is a bioremediation technique where nutrients (like phosphorus or nitrogen) are added to a contaminated environment (like an oil spill) to stimulate the rapid growth of naturally occurring, pollution-eating microbes.
- **Bioaugmentation:** This is the practice of artificially introducing specifically cultured laboratory strains of bacteria into a contaminated site to speed up the degradation of complex pollutants.
- **Additional Facts:**
 - ✓ Urban Mining (E-Waste): Biomining is heavily researched today as an eco-friendly alternative to chemical smelting for extracting precious metals (gold, silver, palladium) from electronic waste (e-waste) circuit boards.
 - ✓ Swachh Bharat Mission Context: While scientifically biomining refers to metal extraction, in Indian civic administration and the Swachh Bharat Mission, "biomining" is the official term used for the bioremediation and mechanical segregation of massive legacy waste in old urban landfills (like Ghazipur in Delhi) to reclaim the land.
 - ✓ Cost and Energy Efficiency: Traditional smelting requires massive furnaces and vast amounts of coal/electricity, producing heavy carbon emissions. Biomining operates at near room temperature and relies entirely on biological processes, drastically reducing the carbon footprint of the mining industry.

Q 123. Ans. B

Explanation:

- In 1796, the English physician Edward Jenner developed the world's first successful vaccine, which was designed to combat smallpox. He made a crucial historical observation that milkmaids who had previously contracted cowpox (a relatively mild disease) were naturally immune to the highly contagious and deadly smallpox virus. He tested this by deliberately inoculating a young boy with material from a cowpox sore, which successfully provided the boy with immunity against smallpox. This revolutionary concept eventually led to the complete global eradication of smallpox, certified by the WHO in 1980.
- **Polio:** The first successful polio vaccine was developed by Jonas Salk in the 1950s.

- **Rabies:** The first vaccine for rabies was pioneered by Louis Pasteur in 1885.
- **Tuberculosis:** The BCG vaccine for tuberculosis was developed by Albert Calmette and Camille Guérin in 1921.

Q 124. Ans. C

Explanation:

- **Xenotransplantation** is the surgical procedure involving the transplantation, implantation, or infusion into a human recipient of live cells, tissues, or organs from a non-human animal source.
- **Autotransplantation:** This involves transplanting tissue from one part of a person's own body to another part (e.g., skin grafts for burns or bone marrow transplants using the patient's own stem cells).
- **Allotransplantation:** This is the most common type of transplant. It involves transplanting organs or tissues from one human to another non-genetically identical human (e.g., a standard kidney or heart transplant from a donor).
- **Isotransplantation:** Also known as syngeneic transplantation, this refers to a transplant between two genetically identical individuals, such as identical (monozygotic) twins.
- **Additional Facts:**
 - ✓ **The Porcine Solution:** Genetically modified pigs are currently the primary candidates for xenotransplantation. Their organs are remarkably similar in size and physiological function to human organs.
 - ✓ **Hyperacute Rejection:** The biggest historical hurdle to xenotransplantation was the human immune system instantly attacking the animal organ. To solve this, scientists use CRISPR gene-editing to remove the alpha-gal sugar gene from pigs (a sugar found in most mammals, but not humans), which normally triggers this violent immune response.
 - ✓ **Zoonotic Disease Risk:** A major ethical and biological concern surrounding xenotransplantation is the risk of "xenozoonosis"—the possibility of transmitting animal viruses, specifically Porcine Endogenous Retroviruses (PERVs), into the broader human population.

Q 125. Ans. C

Explanation:

- **Quantum tunnelling** is a fundamental quantum mechanical phenomenon. It occurs when a

subatomic particle (like an electron) passes through a physical or potential energy barrier. Classically, the particle does not possess enough kinetic energy to surmount this barrier. However, due to its wave-like nature, there is a strict mathematical probability that it will "tunnel" directly through to the other side.

- **Quantum Superposition** is the principle that allows a quantum computer's qubits to represent both 0 and 1 at the exact same time, providing massive parallel processing power.
- **Quantum Entanglement** refers to two or more particles becoming intrinsically linked. The physical measurement of one instantly dictates the state of the other, regardless of spatial distance.
- **Quantum Decoherence** is the primary technological obstacle in quantum computing. It occurs when minor environmental noise (like heat or electromagnetic waves) destroys the quantum state, causing the system to revert to classical binary states and lose its computational advantage.
- **Additional Facts:**
 - ✓ **Stellar Nuclear Fusion:** Quantum tunnelling is the exact physical mechanism that powers the Sun. It allows hydrogen protons to overcome their mutual electrostatic repulsion and fuse together at core temperatures that are classically too low for fusion.
 - ✓ **Modern Electronics:** The phenomenon is actively utilised in everyday consumer electronics. Solid-state drives (SSDs) and USB flash memory drives use quantum tunnelling to erase and rewrite data across insulating oxide layers.
 - ✓ **Scanning Tunnelling Microscope (STM):** This advanced scientific instrument uses the tunnelling of electrons between a sharp metallic tip and a conducting surface to image materials at the individual atomic level.

Q 126. Ans. B

Explanation:

- **Statement 1 is incorrect:** B-lymphocytes are responsible for humoral immunity (antibody-mediated immunity), not cell-mediated immunity. When they encounter a pathogen, they produce an army of specialised proteins called antibodies that circulate in the blood plasma to neutralise the threat. It is actually the Cytotoxic (Killer) T-cells that are responsible for cell-mediated immunity, directly attacking the body's own cells that have become infected by viruses or turned cancerous.

- **Statement 2 is correct:** T-lymphocytes fundamentally differ from B-cells because they never produce antibodies themselves. However, a specific subset called “Helper T-cells” acts as the command centre of the immune response. When they detect an infection, they release chemical signals (cytokines) that actively stimulate and “help” the B-lymphocytes to start producing antibodies. Without Helper T-cells, B-cells cannot effectively mount an antibody response.

Q 127. Ans. B

Explanation:

- **Tokamak** is an experimental machine designed to harness the energy of nuclear fusion. The core of a tokamak is a doughnut-shaped (toroidal) vacuum chamber. Because the plasma (superheated gas) inside must reach temperatures over 100 million degrees Celsius, much hotter than the centre of the Sun, and no solid material can contain it. Therefore, the Tokamak uses incredibly powerful magnetic coils to confine the plasma in mid-air, preventing it from touching the walls.
- **Additional Facts:**
 - ✓ **ITER** (International Thermonuclear Experimental Reactor): This is the world’s largest tokamak, currently under construction in southern France. It is a massive international collaboration involving 35 nations, including India, aiming to prove the feasibility of fusion as a large-scale, safe, and carbon-free energy source.
 - ✓ **The Fourth State of Matter:** To achieve fusion, gases are heated to extreme temperatures until they become plasma. In a plasma, electrons are entirely stripped from the atomic nuclei, making it highly electrically conductive and susceptible to magnetic fields.
 - ✓ **Abundant Fuel:** Unlike nuclear fission (which splits heavy, radioactive elements like Uranium), fusion combines light elements. It primarily uses isotopes of hydrogen: Deuterium (easily extracted from seawater) and Tritium. This means the fuel supply for fusion is virtually inexhaustible and produces no long-lived high-level radioactive waste.

Q 128. Ans. D

Explanation:

- **Kessler Syndrome** (also known as collisional cascading) is a theoretical scenario proposed

by NASA scientist Donald J. Kessler in 1978. It describes a situation where the density of space junk and objects in Low Earth Orbit (LEO) becomes so high that a single collision creates a massive cloud of shrapnel. This new debris then acts as uncontrollable projectiles, striking other satellites and generating even more debris, creating a dangerous, self-sustaining domino effect.

● Additional Facts:

- ✓ **Project NETRA:** This is an early warning system in space developed by ISRO (Indian Space Research Organisation) to detect debris and other hazards to Indian satellites. It stands for Network for space object Tracking and Analysis.
- ✓ **“Graveyard Orbit”:** To prevent the Kessler Syndrome, international space guidelines recommend that when geosynchronous satellites reach the end of their operational life, they should be pushed into a higher “graveyard orbit” safely out of the way of operational traffic.
- ✓ **Anti-Satellite (ASAT) Tests:** India’s “Mission Shakti” in 2019 successfully tested an ASAT weapon. While highly strategic, such kinetic tests are globally controversial precisely because they generate thousands of pieces of new orbital debris.

Q 129. Ans. B

Explanation:

- **Pair 1 is correctly matched:** HELIOS (High Energy Laser with Integrated Optical-dazzler and Surveillance) is a highly advanced laser weapon system developed by Lockheed Martin for the United States. It is primarily integrated into US Navy warships to blind or physically destroy hostile drones and fast-attack boats.
- **Pair 2 is incorrectly matched:** The Peresvet laser complex is a strategic directed energy weapon developed by Russia, not the UK. It is reportedly designed to blind enemy optical reconnaissance satellites. The United Kingdom’s flagship DEW programs are the DragonFire high-power laser and the Radio Frequency Directed Energy Weapon (RFDEW).
- **Pair 3 is correctly matched:** The Silent Hunter (along with the Hurricane-3000) is a prominent laser air-defence system developed by China. It is explicitly designed to counter the growing threat of low-altitude, slow-moving drone swarms by burning through their fuselage.

- **Pair 4 is correctly matched:** The IDD&IS (Integrated Drone Detection and Interdiction System) Mk2A is an indigenous Indian system developed by the DRDO. It is a comprehensive anti-drone umbrella that features “hard kill” (laser-based DEW) capabilities to neutralise aerial threats.
- **Additional Facts:**
 - ✓ The Cost-Per-Shot Advantage: The greatest strategic advantage of DEWs is their economics. Traditional surface-to-air missiles can cost millions of dollars each, creating an unsustainable cost-exchange ratio when shooting down a \$2,000 commercial drone. A laser weapon, however, costs very less worth of electricity per shot and has a virtually limitless “magazine” as long as the generator has fuel.
 - ✓ Limitations of DEWs: Laser weapons currently face significant environmental limitations. Heavy rain, dense fog, dust storms, and atmospheric turbulence can scatter the laser beam, severely reducing its effective range and lethality.
 - ✓ KALI Project: Apart from the IDD&IS, India has historically worked on the KALI (Kilo Ampere Linear Injector) project. Jointly developed by the Bhabha Atomic Research Centre (BARC) and DRDO, it is designed to emit powerful bursts of electromagnetic microwaves to fry the electronic circuitry of incoming missiles.

Q 130. Ans. C

Explanation:

- The yellowish fluid called colostrum, which is secreted by the mother during the initial days of lactation, has very high concentrations of **IgA** antibodies. These antibodies protect the infant’s digestive and respiratory tracts from infections during the early, vulnerable stages of life.
- **IgG:** The most abundant antibody in human blood. It is the only antibody capable of crossing the placenta from mother to fetus to provide immunity before birth.
- **IgE:** This antibody is primarily responsible for triggering allergic reactions (like asthma or hay fever) and protecting against parasitic worm infections.
- **IgM:** This is the largest antibody and is always the “first responder” that the body produces when fighting a new infection.

Q 131. Ans. A

Explanation:

- **Statement 1 is correct:** Ballistic missiles operate essentially like a thrown rock; they are propelled by heavy rocket engines only during the initial boost phase and then rely on gravity and momentum to fall onto their target. Cruise missiles, however, operate more like pilotless aircraft. They are equipped with jet engines (like turbofans or ramjets) that provide continuous thrust for the entirety of their flight.
- **Statement 2 is correct:** Ballistic missiles follow a distinct, high-arching parabolic (sub-orbital) trajectory. Long-range ballistic missiles will completely exit the Earth’s atmosphere into space before re-entering to strike. Cruise missiles, on the other hand, fly Aerodynamically. They stay entirely within the atmosphere and often fly extremely close to the ground (terrain-hugging) to avoid enemy radar detection.
- **Statement 3 is incorrect:** The Agni series (Agni I to Agni V) is India’s premier surface-to-surface ballistic missile, capable of carrying nuclear warheads over intercontinental distances. The BrahMos (jointly developed with Russia) is a supersonic Cruise missile, renowned for its extreme speed and precision manoeuvring.
- **Additional Facts:**
 - ✓ Because ballistic missiles re-enter the atmosphere at hypersonic speeds (often Mach 15-20 for ICBMs), they are incredibly difficult to shoot down in their terminal phase.
 - ✓ Cruise missiles are much slower, but they are difficult to intercept because their low-altitude flight path hides them beneath the “radar horizon” until they are very close to the target.
 - ✓ Ballistic missiles are specifically designed to carry heavier payloads, particularly strategic nuclear warheads, over thousands of kilometres. Cruise missiles generally carry smaller, conventional high-explosive payloads for pinpoint surgical strikes (though some can be nuclear-armed).
 - ✓ Nirbhay Missile: Aside from BrahMos, India has also indigenously developed the ‘Nirbhay’ missile, which is a long-range, subsonic cruise missile designed by the DRDO.

Q 132. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The Iron Dome is a highly renowned, battle-tested mobile air defence

system developed by Rafael Advanced Defence Systems in Israel. It is specifically designed to intercept and destroy short-range rockets and artillery shells fired from close distances.

- **Pair 2 is incorrectly matched:** The Patriot (Phased Array Tracking Radar to Intercept on Target) system is strictly an American-made surface-to-air missile system. Developed primarily by Raytheon, it is the primary high-altitude defence system used by the United States Army and widely exported to allied nations.
- **Pair 3 is correctly matched:** The HQ-9 is China's premier domestic medium-to-long-range, active radar homing surface-to-air missile system. It is heavily analogous to the Russian S-300 and the American Patriot systems, providing comprehensive fleet and territorial air defence for the Chinese military.
- **Pair 4 is incorrectly matched:** The SAMP/T (Surface-to-Air Missile Platform/Terrain) is a purely European air defence architecture. It was jointly developed by France and Italy through the Eurosam consortium. It utilises the Aster 30 missile to intercept aircraft and short-range ballistic missiles.
- **Additional Facts:**
 - ✓ THAAD (Terminal High Altitude Area Defence): Another critical United States system, THAAD is designed to shoot down short, medium, and intermediate-range ballistic missiles in their terminal phase (descent or reentry) using a hit-to-kill approach.
 - ✓ Arrow 3: Co-developed by Israel and the United States, this system handles exoatmospheric interception. It is designed to destroy long-range ballistic missiles while they are still in space, far above the Earth's atmosphere.

Q 133. Ans. B

Explanation:

- HIV belongs to a specialised group of viruses called retroviruses, meaning it carries its genetic information in the form of RNA rather than DNA. Once HIV enters the human body (often initially taking shelter in macrophages), it specifically seeks out, infects, and replicates within Helper T-lymphocytes.
- As the virus multiplies, it destroys these Helper T-cells in the process. Because Helper T-cells are the "command centres" that stimulate B-cells to make antibodies and activate other immune

responses, their drastic depletion leaves the body completely vulnerable. This severe weakening of the immune system is what characterises Acquired Immunodeficiency Syndrome (AIDS), making the patient highly susceptible to opportunistic infections (like tuberculosis or pneumonia) that a healthy immune system would normally fend off easily.

Q 134. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Malaria is not caused by a virus or a bacterium, but by a protozoan parasite of the genus *Plasmodium* (such as *P. falciparum* or *P. vivax*). It is transmitted to humans exclusively through the bites of infected female *Anopheles* mosquitoes, which typically bite between dusk and dawn.
- **Pair 2 is correctly matched:** Dengue is a viral infection caused by the Dengue virus (DENV). It is transmitted primarily by infected female *Aedes aegypti* mosquitoes. Unlike the malaria vector, *Aedes* mosquitoes are predominantly day-biters and breed in clean, stagnant artificial water containers.
- **Pair 3 is incorrectly matched:** Chikungunya is not a bacterial infection. It is a viral disease caused by the Chikungunya virus (CHIKV). However, the second half of the statement is true; it shares the exact same transmitting vector as Dengue, the *Aedes* mosquito. The disease is characterised by an abrupt onset of fever, frequently accompanied by severe, debilitating joint pain.
- **Pair 4 is correctly matched:** Lymphatic Filariasis (commonly referred to as Elephantiasis) is caused by infection with parasitic nematode worms, most commonly *Wuchereria bancrofti*. In India, the primary transmitting vector is the *Culex quinquefasciatus* mosquito. The parasite severely impairs the lymphatic system, leading to the abnormal and painful enlargement of body parts.
- **Additional Facts:**
 - ✓ Acute Encephalitis Syndrome (AES) is locally known as "Chamki Fever" in Bihar. Outbreaks frequently occur in Muzaffarpur, Bihar. While some AES cases are vector-borne (like Japanese Encephalitis, transmitted by *Culex* mosquitoes), the specific Muzaffarpur outbreaks have been strongly linked to a naturally occurring toxin (MCPG) found in unripe litchi fruits consumed by malnourished children.

- ✓ The Government of India has set aggressive public health targets to eliminate these diseases: Lymphatic Filariasis by 2027 and Malaria by 2030.
- ✓ Platelet Drop: A defining clinical characteristic of severe Dengue fever (often called “break-bone fever” due to intense joint pain) is a rapid, dangerous drop in the patient’s blood platelet count (thrombocytopenia).
- ✓ Quinine: The oldest and most famous anti-malarial drug, Quinine, is naturally extracted from the bark of the Cinchona tree.

Q 135. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The Udaipur Solar Observatory is located on an island in Fateh Sagar Lake in Rajasthan. It is managed by the Physical Research Laboratory. It is considered one of the best solar observing sites in Asia.
- **Pair 2 is incorrectly matched:** The Kodaikanal Solar Observatory is located in the Palani Hills of Tamil Nadu, not Kerala. It is historically significant and is operated by the Indian Institute of Astrophysics.
- **Pair 3 is correctly matched:** The Gauribidanur Radio Observatory is located near Bengaluru in Karnataka. It is jointly operated by the Raman Research Institute and the Indian Institute of Astrophysics. It operates a dedicated radio heliograph to monitor the solar corona.
- **Pair 4 is incorrectly matched:** The Aryabhata Research Institute of Observational Sciences (ARIES) is located in Nainital, Uttarakhand, not Himachal Pradesh. It is an autonomous institute that conducts extensive research in solar physics and atmospheric sciences.
- **Additional Facts:**
 - ✓ Aditya-L1 Mission: This is India’s first dedicated space-based solar observatory. It was successfully launched by the Indian Space Research Organisation (ISRO) using the PSLV-C57 rocket.
 - ✓ Lagrange Point 1: The Aditya-L1 spacecraft is placed in a halo orbit around the Sun-Earth Lagrange point 1 (L1). This strategic position provides a continuous, uninterrupted view of the Sun without any occultation or eclipses.
 - ✓ MAST Telescope: The Multi Application Solar Telescope (MAST) is a prominent facility operationalized at the Udaipur Solar Observatory. It provides highly detailed 3D velocity and magnetic field models of active solar flares.

Q 136. Ans. A

Explanation:

- ✓ **Pair 1 is correctly matched:** mRNA vaccines (like the Pfizer or Moderna COVID-19 vaccines) use a tiny piece of messenger RNA wrapped in a protective lipid nanoparticle to enter host cells. Once inside, they instruct the cells to temporarily build a specific viral protein (like the spike protein), which strongly stimulates both B-cells (humoral immunity) and T-cells (cellular immunity).
- ✓ **Pair 2 is incorrectly matched:** Live-attenuated vaccines (like MMR or BCG) use a living, but severely weakened form of the pathogen that closely mimics a natural infection to create a very strong and long-lasting cellular immune response. The mechanism described in the pair, using a chemically killed pathogen, actually describes an Inactivated vaccine (like Covaxin or the Salk polio vaccine).
- ✓ **Pair 3 is correctly matched:** Viral vector vaccines (like Covishield/AstraZeneca or Sputnik V) take a completely different, harmless virus (often an adenovirus) and modify it to act as a Trojan horse. This “vector” delivers the genetic code of the target pathogen into our cells to trigger a robust cellular and humoral immune response.
- ✓ **Pair 4 is incorrectly matched:** Toxoid vaccines (like those for Tetanus or Diphtheria) use an inactivated version of the harmful toxin produced by the bacteria, rather than targeting the bacteria itself. The mechanism described in the pair, using an isolated protein or sugar capsule, actually describes a Subunit or Conjugate vaccine (like the Hepatitis B vaccine).

Q 137. Ans. B

Explanation:

- **Option A is incorrect:** While satellites do transmit data, they do not operate completely independently of the ground. They absolutely require terrestrial gateway stations to interface with the global fibre-optic internet backbone. They cannot just exclusively bounce signals between individual user terminals.
- **Option B is correct:** Satellite broadband provides direct wireless internet connectivity. It uses communication satellites actively orbiting the Earth. The physical infrastructure fundamentally

requires a space segment composed of satellite constellations. It also strictly necessitates a ground segment featuring gateway stations and network control centres to connect to the wider global internet.

- **Option C is incorrect:** Bouncing radio waves off the ionosphere is a standard propagation technique for long-distance amateur (ham) radio. This specific atmospheric method is physically incapable of supporting the high-bandwidth frequencies required for broadband internet.
- **Option D is incorrect:** Modern Low Earth Orbit (LEO) satellites do utilise inter-satellite laser links. This reduces global routing latency by passing data across the constellation. However, they cannot completely bypass ground-based infrastructure. The data must eventually be beamed down to a terrestrial gateway to reach standard terrestrial servers.
- **Additional Facts:**
 - ✓ Latency: Satellite internet typically suffers from higher latency compared to terrestrial wired connections. This delay occurs because data must physically travel thousands of kilometres up to the space segment and back down to the user.
 - ✓ LEO Megaconstellations: Modern projects rely heavily on Low Earth Orbit (LEO) under 2,000 km altitude. This drastically reduces the signal latency that historically plagued older Geostationary (GEO) internet satellites.
 - ✓ The socio-economic objective of these systems is universal connectivity. They are uniquely capable of providing reliable, high-speed broadband to difficult mountainous or rural terrains where laying underground optical fibre is economically unviable.
 - ✓ Key Global Projects: Prominent operational and planned megaconstellations include SpaceX's Starlink, Amazon's Project Kuiper, Eutelsat's OneWeb, and China's Qianfan constellation.

Q 138. Ans. B

Explanation:

- **Sun-Synchronous Orbit (SSO)** is a specific type of near-polar orbit. The satellite's orbital plane rotates (precesses) at the same rate at which the Earth revolves around the Sun. Consequently, the satellite maintains a constant relationship with the Sun, ensuring it passes over any specific location on the Earth's surface at the same local solar time every day. This provides consistent lighting conditions,

which are essential for Earth-observation and imaging missions like NISAR.

- **Geostationary Orbit (GEO):** While ISRO's GSLV rocket is typically used for placement in Geosynchronous Transfer Orbit (GTO) to reach GEO, NISAR marks the first time a GSLV was used to place a satellite into a Sun-Synchronous Orbit.
- **Molniya orbits** are highly inclined, highly elliptical orbits designed to provide consistent satellite coverage over polar and high-latitude regions where geostationary satellites are less effective.
- **Prograde Orbit:** While many launches use the Earth's rotation for a "boost," Sun-Synchronous orbits are often "retrograde" (slightly opposite to Earth's rotation) to achieve the specific nodal precession required to stay in sync with the Sun.
- **Additional Facts:**
 - ✓ NISAR was launched in July 2025 using ISRO's GSLV-F16 from the Satish Dhawan Space Centre, Sriharikota.
 - ✓ Dual-Band Innovation: It is the world's first Earth observation satellite to integrate both L-band (provided by NASA) and S-band (provided by ISRO) Synthetic Aperture Radar on a single platform.
 - ✓ Large-Scale Deployment: Following the launch, the mission successfully deployed a massive 12-metre diameter common unfurlable reflector antenna.
 - ✓ Advanced Imaging: It utilises SweepSAR technology to produce high-resolution, wide-swath images. The radar works by sending out energy pulses and recording the reflection back from Earth's surface.
 - ✓ Monitoring Capability: With a mission life of 5 years, it provides high-resolution data every 12 days (repeat cycle) to monitor land and ice changes across the globe.
 - ✓ Global Collaboration: The mission follows an Open-Data Policy, making its findings freely available to the global scientific community.

Q 139. Ans. C

Explanation:

- **Anthrax** is a severe, often fatal, infectious disease caused by the rod-shaped, Gram-positive bacterium **Bacillus anthracis**. It primarily affects herbivorous livestock (like sheep and cattle) but can be transmitted to humans through contact with infected animals or their products, causing severe respiratory, gastrointestinal, or skin infections.

- **Bacillus cereus:** Primarily causes foodborne illnesses (food poisoning), famously associated with reheated fried rice.
- **Yersinia pestis:** The causative agent of the Plague (Bubonic, Septicemic, and Pneumonic), historically known as the Black Death.
- **Clostridium perfringens:** A common cause of severe food poisoning and the primary agent responsible for gas gangrene in wounds.
- **Mycobacterium tuberculosis:** The causative agent of Tuberculosis (TB).
- **Clostridium tetani:** The causative agent of Tetanus (Lockjaw).
- **Vibrio cholerae:** The causative agent of Cholera.

Q 140. Ans. D

Explanation:

- Polysaccharides are complex carbohydrates (macronutrients) like Starch (found in plants) and Glycogen (the storage form in animals).
- Selenium and Zinc are trace minerals (micronutrients) required by the body in very small amounts (milligrams or micrograms).
- Folic Acid (B9) and Biotin (B7) belong to the B-complex group, which are water-soluble vitamins.
- Lysine and Methionine are essential amino acids, which are the building blocks of proteins (macronutrients) that the body cannot synthesise on its own.
- **Additional facts:**
 - ✓ **Dietary Fibre:** Often considered a “non-nutrient” carbohydrate, it is a polysaccharide (cellulose) that humans cannot digest but is essential for gut health.
 - ✓ **Major vs. Trace Minerals:** Minerals like Calcium and Magnesium are “Major Minerals” (needed in >100mg/day), whereas Iron and Iodine are “Trace Minerals.”
 - ✓ **Fat-Soluble Vitamins:** Unlike the water-soluble examples above, Vitamins A, D, E, and K require dietary fat for absorption and can be stored in the liver.
 - ✓ **Complete Proteins:** Animal sources usually provide all essential amino acids, while plant sources (except soy and quinoa) are often “incomplete” and must be combined (e.g., rice and dal).

Q 141. Ans. A

Explanation:

- The nanometer (nm) rating historically refers to the physical dimensions of individual transistors on a semiconductor chip. A lower nm rating

signifies that the transistors are physically smaller. This allows manufacturers to pack billions more transistors onto a single silicon die. Smaller transistors require lower electrical voltage to switch states. The reduced physical distance also means electrons travel faster between components. This directly results in higher processing speeds and significantly enhanced battery efficiency.

● Additional Facts:

- ✓ **Moore’s Law:** An empirical observation formulating that the number of transistors on a microchip doubles approximately every two years. This economic and technological driver pushes the industry toward increasingly smaller nodes (e.g., 5nm, 3nm).
- ✓ **Quantum Tunnelling:** As transistors shrink toward the 2nm threshold, the insulating barriers become perilously thin. Electrons can spontaneously “tunnel” through these barriers. This causes severe electrical leakage and heat generation.
- ✓ **Transistor Architecture:** To combat quantum leakage, the industry shifted from flat planar transistors to complex 3D structures. These include Fin Field-Effect Transistors (FinFET) and Gate-All-Around (GAA) architectures. They wrap the gate around the electron channel to prevent leakage.
- ✓ **India Semiconductor Mission (ISM):** A flagship initiative launched by the Ministry of Electronics and IT (MeitY). It operates with a massive ₹76,000 crore outlay. It aims to establish a robust, indigenous semiconductor manufacturing and design ecosystem to ensure technological sovereignty.

Q 142. Ans. D

Explanation:

- Kala-azar, a severe public health concern in states like Bihar, is a parasitic disease transmitted through the bite of infected female **phlebotomine sandflies**.
- The **female Anopheles mosquito** is the specific vector for the Plasmodium parasite, which causes Malaria.
- The **Aedes aegypti mosquito** is the vector for viral diseases such as Dengue, Chikungunya, and Zika. It is famously known for breeding in clean, stagnant water and biting during the daytime.
- The **Culex mosquito** is primarily responsible for transmitting the pathogen that causes Lymphatic Filariasis (Elephantiasis) as well as the Japanese Encephalitis virus.

Q 143. Ans. B**Explanation:**

- *Haemophilus influenzae* is the bacterial causative agent of pneumonia, along with *Streptococcus pneumoniae*. The infection inflames the alveoli, filling them with fluid and impairing gaseous exchange.
- *Salmonella typhi* causes Typhoid fever, a systemic infection primarily affecting the gastrointestinal tract.
- *Mycobacterium leprae* causes Leprosy (Hansen's disease), a chronic disease affecting the skin and peripheral nerves.
- *Wuchereria bancrofti* is a parasitic nematode responsible for Filariasis (Elephantiasis), transmitted by mosquitoes and causing lymphatic obstruction.

Q 144. Ans. C**Explanation:**

- Modern undersea cables are fibre-optic systems. They transmit data encoded as rapid pulses of laser light. This light travels through thin strands of highly purified silica glass. It remains confined within the central core due to the physical phenomenon of total internal reflection. Because the light signal gradually attenuates over transoceanic distances, devices called optical repeaters are spliced into the cable to periodically amplify the light.
- Coaxial submarine cables relied on heavy copper wires to transmit electrical analogue signals. These were superseded by fibre optics in the late 1980s. Copper suffers from severe resistance and signal loss over long distances, offering vastly lower data bandwidth compared to light.
- Microwaves are high-frequency electromagnetic waves used extensively for terrestrial line-of-sight links and satellite communications. However, they suffer from extreme absorption and rapid attenuation when exposed to liquid water. They are physically unsuited for underwater data transmission.
- Acoustic wave transmission is the mechanism behind sonar systems. It is used for communication between submarines or deep-sea drones. While sound travels efficiently through water, acoustic signals possess an extremely low bandwidth. They are mathematically incapable of carrying the massive, high-speed data loads required by the global internet.

Additional Facts:

- ✓ **Total Internal Reflection (TIR):** This critical optical phenomenon occurs when a light ray hits the boundary between an optically denser medium (the fiber core) and a rarer medium (the cladding) at an angle greater than the critical angle. This forces 100% of the light to reflect back inward, preventing signal leakage.
- ✓ **Optical Repeaters:** These are pressurised, cylindrical housings spaced roughly every 50 to 100 kilometres along the seabed. Modern repeaters utilise Erbium-Doped Fibre Amplifiers (EDFAs) to optically boost the fading laser signal without converting it back into a slow electrical signal.
- ✓ **Power Conductor:** While the actual data travels purely as photons (light), the submarine cable still contains a copper or aluminium conductive sheath. This sheath carries high-voltage direct current (HVDC) from the shore stations to physically power the repeaters lying on the ocean floor.
- ✓ **Strategic Importance:** Over 95% of all international data, financial transactions, and internet traffic is routed through these submarine cables rather than satellites, making them highly vulnerable critical infrastructure.

Q 145. Ans. A**Explanation:**

- **Electroencephalogram (EEG)** records the electrical activity of the brain. It is the primary method used in non-invasive Brain-Computer Interfaces. It detects voltage fluctuations resulting from ionic current within the neurons of the brain through electrodes placed externally on the scalp.
- **Electrocardiogram (ECG)** measures the electrical activity of the heart. It is used to detect cardiac abnormalities and arrhythmias, not to decode cognitive intentions or neural motor commands.
- **Electromyogram (EMG)** records the electrical activity produced by skeletal muscles. While highly useful in advanced prosthetics, it measures peripheral muscle activation rather than central brain activity.
- **Electrooculogram (EOG)** measures the corneo-retinal standing potential that exists between the front and the back of the human eye. It is utilised to track eye movements, not the localised activity of the cerebral cortex.
- **Additional Facts:**

- ✓ **Mechanism of Action:** BCIs completely bypass traditional neuromuscular pathways. They directly translate neural signals into actionable algorithmic commands to control external devices, such as robotic limbs, wheelchairs, or speech synthesisers.
- ✓ **Invasive vs. Non-Invasive:** Invasive BCIs require the surgical implantation of microelectrode arrays directly into the brain's grey matter for high-fidelity signals. Non-invasive BCIs (like EEG) are safer but suffer from severe signal attenuation and noise caused by the human skull.
- ✓ **Neuroplasticity:** The successful use of a BCI often relies on neuroplasticity. The user's brain must adapt and learn to generate specific, recognisable neural patterns that the machine learning algorithm can consistently decode.
- ✓ **Clinical Applications:** The technology is heavily researched for neuro-rehabilitation. It is primarily targeted at restoring communication and mobility for patients suffering from severe motor impairments, such as Amyotrophic Lateral Sclerosis (ALS) or complete spinal cord injuries.

Q 146. Ans. A

Explanation:

- **Option A is correct:** Anorexia nervosa is a psychiatric eating disorder fundamentally marked by restricted eating, severe weight loss, and an intense fear of gaining weight. Individuals with this condition often have a distorted body image.
- **Option B is incorrect:** This statement describes the clinical presentation of Bulimia Nervosa, where patients engage in a binge-purge cycle.
- **Option C is incorrect:** This statement describes Binge Eating Disorder, which lacks the compensatory purging behaviours seen in bulimia.
- **Option D is incorrect:** This describes a physiological endocrine disorder (Hyperthyroidism).
- **Additional facts:**
 - ✓ **Pica:** A persistent craving and consumption of non-food, non-nutritive substances (e.g., clay, dirt, chalk, ice). It is often associated with iron-deficiency anaemia or developmental disorders.
 - ✓ **Rumination Disorder:** Repeated, involuntary regurgitation of recently ingested food, which is then re-chewed, re-swallowed, or spit out.
 - ✓ **Avoidant/Restrictive Food Intake Disorder**

(ARFID): Highly selective eating habits or lack of interest in eating, leading to significant nutritional deficiency, but without the distorted body image or fear of weight gain seen in anorexia.

Q 147. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Insomnia is clinically defined not just by a lack of sleep, but by the difficulty in falling or staying asleep even when the environment is conducive to rest.
- **Pair 2 is correctly matched:** Melatonin is often called the "hormone of darkness." Its secretion increases in the evening, helping to synchronise the body's internal circadian rhythm with the external environment.
- **Pair 3 is correctly matched:** Dyspnoea is a symptom where an individual feels they cannot get enough air. It is a critical diagnostic sign for asthma, heart failure, and pneumonia.
- **Pair 4 is correctly matched:** Anosmia is the medical term for the loss of smell. While it can be caused by nasal congestion or polyps, it gained global prominence as a specific neurological/respiratory symptom of SARS-CoV-2.
- **Additional facts:**
 - ✓ **Sleep hygiene:** A set of behavioural and environmental recommendations (like avoiding blue light before bed) designed to promote better sleep quality for those with insomnia.
 - ✓ **Apnoea:** A sleep disorder where breathing repeatedly stops and starts; unlike insomnia, it is a physical obstruction or neurological lapse in breathing.
 - ✓ **Ageusia:** Often paired with Anosmia in clinical settings, this refers to the loss of the sense of taste.
 - ✓ **Cyanosis:** A bluish discolouration of the skin or mucous membranes resulting from poor circulation or inadequate oxygenation of the blood (often a late-stage sign following prolonged Dyspnoea).

Q 148. Ans. B

Explanation:

- A biochemical marker (or biomarker) is a naturally occurring, measurable substance, such as a specific protein, enzyme, hormone, or genetic sequence, found in blood, bodily fluids, or tissues. Its

presence or abnormal concentration serves as an objective indicator of a normal biological process, a pathogenic disease, or a pharmacological response to a therapeutic intervention.

- **Additional facts:**

- ✓ **Diagnostic utility:** They are the foundation of preventative and diagnostic medicine. They allow physicians to detect diseases internally before physical symptoms ever appear on the outside.
- ✓ **Examples:** Blood Glucose/HbA1c: Biomarkers for Diabetes.
- ✓ **Troponin:** Biomarker for myocardial infarction (heart attack).
- ✓ **C-Reactive Protein (CRP):** Biomarker for systemic inflammation.
- ✓ **Liquid Biopsy:** A modern S&T application where cancer biomarkers (like circulating tumour DNA) can be detected through a simple, non-invasive blood test instead of surgically removing organ tissue.

Q 149. Ans. B

Explanation:

- **Acute Herpetic Gingivostomatitis** is a common manifestation of a primary HSV-1 infection, often seen in children, resulting in inflammation of the gums and mouth with fluid-filled blisters.
- **Carcinoma** refers to a type of cancer; while some viruses (like HPV) are linked to cancer, HSV is primarily known for vesicular lesions.
- **Rheumatism** refers to inflammation and pain in the joints or muscles, which is not a characteristic symptom of an HSV infection.
- **Hepatitis:** While severe disseminated herpes can occasionally affect the liver in immunocompromised individuals, Hepatitis is primarily caused by specific Hepatitis viruses (A, B, C, etc.).
- **Additional Facts:**
 - ✓ **Herpes Labialis:** Commonly known as “cold sores” or “fever blisters,” these typically appear around the lips and are the most frequent sign of HSV-1 reactivation.

- ✓ **Genital Herpes:** Characterised by painful sores and ulcers in the genital and anal areas, most frequently caused by HSV-2.
- ✓ **Herpetic Keratitis:** An infection of the eye that can cause scarring of the cornea and is a leading cause of corneal blindness worldwide.
- ✓ **Neonatal Herpes:** A serious condition where the virus is transmitted from mother to newborn during childbirth, potentially leading to neurological disability or death.

Q 150. Ans. A

Explanation:

- **Iodine - Goitre:** Iodine is essential for the synthesis of thyroid hormones; its deficiency causes the thyroid gland to enlarge, a condition known as Goitre.
- **Vitamin D - Osteomalacia:** While Vitamin D deficiency causes Rickets in children, it leads to Osteomalacia (softening of bones) in adults.
- **Niacin (Vitamin B3) - Pellagra:** A deficiency of Niacin leads to Pellagra, famously characterised by the “3 Ds”: Dermatitis, Diarrhoea, and Dementia.
- **Vitamin A - Xerophthalmia:** A prolonged deficiency of Vitamin A leads to Xerophthalmia, where the eyes fail to produce tears, eventually leading to corneal damage and blindness.
- **Additional facts:**
 - ✓ **Vitamin B12 (Cyanocobalamin):** Essential for RBC formation and DNA synthesis; deficiency causes Pernicious Anaemia.
 - ✓ **Vitamin C (Ascorbic Acid):** Critical for collagen synthesis and wound healing; deficiency leads to Scurvy (bleeding gums).
 - ✓ **Vitamin K:** Necessary for the synthesis of proteins required for blood clotting; deficiency leads to excessive bleeding from even minor injuries.
 - ✓ **Iron:** A core component of Hemoglobin; deficiency is the leading cause of Anemia globally, particularly among women and children.

73rd BPSC Foundation Batch

Prelims + Mains + Interview

BATCH 9

हिंदी माध्यम

📍 Musallahpur

ONLINE & OFFLINE

STARTS FROM 9th APRIL 2026

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026

BATCH 6

Bilingual

BATCH 1

English Medium

📍 Boring Road

ONLINE & OFFLINE

STARTS FROM 21st APRIL 2026



Patna

Musallahpur

KGS Campus,
Near Sai Mandir,
Musallahpur Hat,
Patna, Bihar-800006

Boring Road

2nd Floor, Kumar
Tower, Boring Road
Chauraha, Patna,
Bihar-800001

Naya Tola

KP Yadav
Compound Kazipur,
Naya Tola, Patna,
Bihar-800004